Rational Voting in Swiss Single-Seat Cantons: The Impact of Candidate Numbers and Ideological Positions on Electoral Outcomes — Code Journal

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Table of contents

1 Setup

This code was used throughout my analysis process. To make the process transparent and understandable, explanatory comments are provided either above each code chunk or alongside the code itself, depending on the length and complexity of the code. Due to the agreement with smartvote, I am not able to release the full dataset. Nevertheless, this code journal should make it possible to understand and trace how the data was prepared and analysed.

1.1 Packages

These Packages were used in the process.

```
library(scales)
library(gt)
library(tidyverse)
library(here)
library(janitor)
library(lubridate)
library(glue)
library(broom)
library(modelsummary)
library(ggrepel)
library(patchwork)
library(knitr)
library(kableExtra)
library(writexl)
library(readxl)
library(ggplot2)
library(ggridges)
library(dplyr)
library(stringr)
library(tidyr)
library(sandwich)
library(tibble)
library(webshot2)
library(lmtest)
```

1.2 Global options

```r
knitr::opts_chunk$set(
  fig.align = "center",
  fig.width = 8,
  fig.height = 5,
  dpi = 300
)
```

```r
# ====================================================
# Helper functions
# ====================================================

na_excel <- function(x) {
  # Turns common Excel error codes and empty strings into NA
  x <- as.character(x)
  x <- stringr::str_squish(x)
  x <- dplyr::na_if(x, "#NV")
  x <- dplyr::na_if(x, "#DIV/0!")
  x <- dplyr::na_if(x, "")
  x
}

chr_num <- function(x) {
  # Safe numeric conversion for character and already-numeric columns
  x <- na_excel(x)
  suppressWarnings(as.numeric(x))
}

pct_to_num <- function(x) {
  # Converts percentages such as "96%" into 0.96
  x <- na_excel(x)
  x <- stringr::str_replace_all(x, "%", "")
  suppressWarnings(as.numeric(x) / 100)
}
```

1.3 Project Structure

The project is organised into separate folders for raw data, cleaned data, scripts, outputs, and documentation. The folder data/raw/ contains the merged candidate dataframe as well as additional information extracted from the RIS analysis that is used during the data preparation process. Cleaned datasets are stored in data/clean/, while all generated tables, figures, and model outputs are saved in output/figures or output/tables.

```r
paths <- list(
  raw   = here("data", "raw"),
  clean = here("data", "clean"),
  figs  = here("output", "figures"),
  tabs  = here("output", "tables")
)

dir.create(paths$raw,   recursive = TRUE, showWarnings = FALSE)
dir.create(paths$clean, recursive = TRUE, showWarnings = FALSE)
dir.create(paths$figs,  recursive = TRUE, showWarnings = FALSE)
```

2 Data Import

This section imports the merged candidate-level dataframe used for the analysis. The dataframe combines national and cantonal National Council and Council of States election data with RIS scores obtained from the PCA analysis based on the smartvote data archive.

In addition to the smartvote data , information from cantonal and national election publications was used. I used the national ARchive data as a basis and merged the Infomrations with additional insights and further details regarding Candidates from their cantonal archives using excel.

——————————————**SRF Election Results Insights**——————————————

2019: https://www.srf.ch/news/schweiz/wer-hat-die-wahl-geschafft-resultate-zu-allen-kandidierenden

2023: https://www.srf.ch/news/wahlen-2023-resultate

——————————————**National Election-archive——-**——————————————

NC

2023: https://www.bk.admin.ch/bk/de/home/politische-rechte/nationalratswahlen/nationalratswahlen-2023/Wahlresultate23.html / https://www.wahlen.admin.ch/de/ch/

2019: https://www.bk.admin.ch/bk/de/home/politische-rechte/nationalratswahlen/nationalratswahlen-2019/Wahlresultate19.html

2015: https://www.bk.admin.ch/ch/d/nrw/nrw15/list/kt_index.html

2011: https://www.bk.admin.ch/ch/d/nrw/nrw11/list/kt_index.html

2007: https://www.bk.admin.ch/ch/d/nrw/nrw07/list/kt_index.html

CS

2023: https://www.wahlen.admin.ch/de/ch/standerat/

2019: https://www.bfs.admin.ch/bfs/de/home/statistiken/politik/wahlen/eidgenoessische-wahlen/2019/resultate-staenderat.html

2015: http://www.politik-stat.ch/srw2015KT_de.html

2011: http://www.politik-stat.ch/srw2011KT_de.html

2007: http://www.politik-stat.ch/srw2007KT_de.html

—————————————————**Obwalden**——————————————————

NC

2023: https://www.ow.ch/*docn/357508/OW-%231528908-v1-2023_10_22_*-_Wahlprotokoll_NR_Excel.pdf

2019: https://www.ow.ch/_docn/191755/Ergebnis_NR-Wahl.pdf

2015: https://www.ow.ch/_docn/72251/Wahlprotokoll_NR.pdf

2011: https://www.ow.ch/_docn/29559/Ergebnis_Wahlen_und_Abstimmung_vom_23.10.2011.pdf

2007: https://www.ow.ch/_docn/11382/Wahlergebnis%20Nationalratswahl%20vom%2021.%20Oktober%202007.pdf

CS

2023: -

2019: -

2015: https://www.ow.ch/_docn/72271/Wahlprotokoll_SR.pdf

2011: https://www.ow.ch/_docn/29559/Ergebnis_Wahlen_und_Abstimmung_vom_23.10.2011.pdf

2007: -

—————————————————**Nidwalden**——————————————————

NC

2023: https://www.nw.ch/_docn/357496/Auswertung_Nationalrat_2023_NW.pdf

2019: https://www.nw.ch/_docn/191731/Schlussergebnis_Nationalratswahlen_2019.pdf

2015: https://www.nw.ch/_docn/72275/AUSWERTUNG_Nationalrat_2015_NW_final.pdf

2011: https://www.nw.ch/_docn/51910/AUSWERTUNG_2011_NW.pdf

2007: https://www.nw.ch/_docn/85417/10-21-2007_Stille_Wahl_Nationalrat.pdf

CS

2023: https://www.nw.ch/_docn/357499/Auswertung_Ständerat_2023_NW.pdf

2019: -

2015: https://www.nw.ch/_docn/72276/AUSWERTUNG_Staenderat_2015_NW_final.pdf

2011:-

2007: https://www.nw.ch/_docn/85418/10-21-2007_Stille_Wahl_Standerat.pdf

Appenzell Ausserrhoden

NC and CS

2023: https://ar.ch/fileadmin/user_upload/Kantonskanzlei/Rechtsdienst/Volksrechte/Abstimmungen_Wahlen/2023/Schlussergebnisse_22._Oktober_2023.pdf

2019: https://ar.ch/fileadmin/user_upload/Kantonskanzlei/Rechtsdienst/Volksrechte/Abstimmungen_Wahlen/2019/Schlussergebnisse_20._Oktober_2019.pdf

2015: https://ar.ch/fileadmin/user_upload/Kantonskanzlei/Rechtsdienst/Volksrechte/Abstimmungen_Wahlen/2015/Schlussergebnisse_18._Oktober_2015_*Auszug_Abl._Nr._43*.pdf

2011: https://query-staatsarchiv.ar.ch/Dateien/42/D214223.pdf

2007: https://query-staatsarchiv.ar.ch/Dateien/51/D259143.pdf

Glarus

NC

2023: https://www.gl.ch/public/upload/assets/51298/Ergebnis%20Nationalratswahl%20Kanton%20Glarus%202023.pdf?fp=1

2019: https://www.gl.ch/public/upload/assets/23632/191020_Ergebnisse%20Nationalratswahl.pdf?fp=1

2015: https://www.gl.ch/public/upload/assets/3502/151018_Nationalratswahl_2015.pdf?fp=1

2011: https://www.gl.ch/public/upload/assets/2156/Nationalratswahl_23_10_2011.pdf?fp=1

2007:https://www.gl.ch/public/upload/assets/1450/Nationalratswahl_21_10_07.pdf?fp=1

Uri

NC

2023: https://www.ur.ch/_docn/357526/StatistikNR-Wahl_2023.pdf

2019:https://www.ur.ch/_docn/191737/NR-Wahl_Sonntag.pdf

2015: https://www.ur.ch/_docn/72255/NR-Wahl.pdf

2011: https://www.ur.ch/_docn/33724/tabelleNR2011.pdf

2007: https://www.ur.ch/_docn/35339/tabellennrsonntag.pdf

Basel-Stadt

CS

2023: https://media.bs.ch/original_file/2530e7d43239731af6b3eed8c2c73e00d68a6e00/w-a-2023-10-22-schlussresultat-sr-v1-1.pdf

2019: https://media.bs.ch/original_file/8813e097100ce37197bfaf1598e60f344b276030/w-a-2019-10-20-rr-schlussresultat.pdf

2015: https://media.bs.ch/original_file/3ec3e9d884ce577ded5039adffb3234d4accc7e3/w-a-2015-10-18-schlussresultat-sr.pdf

2011: https://media.bs.ch/original_file/5383a7549c38db53513a76acc19ef9029801e8d6/w-a-2011-10-23-schlussresulta t-sr.pdf

2007: https://www.bs.ch/medienmitteilungen/staatskanzlei/2007-anita-fetz-als-baselstaedtisches-mitglied-den-staenderat-gewaehlt

Basel-Land

CS

2023: https://abstimmungen.bl.ch/election/staenderatswahlen-2023/pdf

2019: https://abstimmungen.bl.ch/election/staenderatswahlen-2019/pdf

2015: https://abstimmungen.bl.ch/election/staenderatswahlen-2015/pdf

2011: https://abstimmungen.bl.ch/election/staenderatswahlen-2011/pdf

2007: https://abstimmungen.bl.ch/election/staenderatswahlen-2007/pdf

Summary

SRF

2019: https://www.srf.ch/news/schweiz/wer-hat-die-wahl-geschafft-resultate-zu-allen-kandidierenden

2023: https://www.srf.ch/news/wahlen-2023-resultate

National Council Archive [1995-2023]: https://www.bk.admin.ch/bk/de/home/politische-rechte/nationalratswahlen.html

Council of States Archive [2003-2015]: http://www.politik-stat.ch/

Council of States 2019: https://www.bfs.admin.ch/bfs/de/home/statistiken/politik/wahlen/eidgenoessische-wahlen/2019/resultate-staenderat.html

Results Council of States and National Council 2023: https://www.wahlen.admin.ch/de/ch/

Nidwalden: https://www.nw.ch/abstimmungen

Obwalden: https://www.ow.ch/dienstleistungen/2045

Glarus: https://www.gl.ch/verwaltung/staatskanzlei/wahlen-und-abstimmungen/archiv.html/1287

Uri: https://www.ur.ch/abstimmungen

Appenzell Ausserrhoden: https://ar.ch/verwaltung/kantonskanzlei/rechtsdienst/politische-rechte/wahlen-abstimmungen/archiv/ and https://query-staatsarchiv.ar.ch/detail.aspx?Id=581034

Basel-Stadt: https://www.bs.ch/regi2019erungsrat/staatskanzlei/politische-rechte/wahlen-und-abstimmungen/wahl-und-abstimmungsresultate-archiv

Basel-Land: https://www.baselland.ch/politik-und-behorden/besondere-behoerden/landeskanzlei/politische_rechte/wahlen/staenderatswahlen/archiv-staenderatswahlen

```r
cand <- read_delim(
  here("data", "raw", "candidates_clean.csv"),
  delim = ";",
  locale = locale(decimal_mark = "."),
  na = c("", "NA", "#NV", "#DIV/0!"),
  show_col_types = FALSE
) |>
  clean_names()
```

2.1 Data cleaning

```r
# --- Clean & standardize dataset ---

cand <- cand |>
  mutate(
    # standard core identifiers
    name = stringr::str_squish(name),
    id_ba = stringr::str_squish(id_ba),
    year = as.integer(elec_year),
    canton = stringr::str_squish(elec_canton),

    chamber = case_when(
```

```r
    stringr::str_detect(elec_council,
                        regex("National", ignore_case = TRUE)) ~ "NR",
    stringr::str_detect(elec_council,
                        regex("Council of States", ignore_case = TRUE)) ~ "SR",
    TRUE ~ NA_character_
  ),

  party = stringr::str_squish(pol_party),

  # election outcome
  win = case_when(
    stringr::str_to_lower(elec_result) == "win"  ~ 1L,
    stringr::str_to_lower(elec_result) == "loss" ~ 0L,
    TRUE ~ NA_integer_
  ),

  # numeric conversions
  total_votes = chr_num(total_votes),
  personal_votes = chr_num(personal_votes),
  personal_votes_share = pct_to_num(personal_votes_percentage),

  # RIS: "#NV" / empty values become NA
  ris = chr_num(ris),
  ris = if_else(ris == 0, NA_real_, ris),

  # smartvote yes/no to 1/0
  smartvote = case_when(
    stringr::str_to_lower(stringr::str_squish(smartvote)) == "yes" ~ 1L,
    stringr::str_to_lower(stringr::str_squish(smartvote)) == "no"  ~ 0L,
    TRUE ~ NA_integer_
  ),

  # smartvote policy dimensions
  foreign_policy = chr_num(foreign_policy),
  liberal_economic_policy = chr_num(liberal_economic_policy),
  restrictive_financial_policy = chr_num(restrictive_financial_policy),
  law_order = chr_num(law_order),
  restrictive_migration_policy = chr_num(restrictive_migration_policy),
  environmental_effort = chr_num(environmental_effort),
  social_state = chr_num(social_state),
  liberal_society = chr_num(liberal_society)
) |>
```

```r
  mutate(
    # election id for grouping
    election_id = stringr::str_c(canton, year, chamber, sep = "_"),

    # candidate vote share
    # If personal_votes_percentage is available, prefer it.
    vote_share = case_when(
      !is.na(personal_votes_share) ~ personal_votes_share,
      !is.na(personal_votes) &
        !is.na(total_votes) & total_votes > 0
      ~ personal_votes / total_votes,
      TRUE ~ NA_real_
    )
  )
```

```r
cand <- cand |>
  mutate(
    others = if_else(
      is.na(gender),
      1L,  # Sammelvariable / Others
      0L   # real candidate
    )
  )
cand <- cand |>
  filter(!is.na(name))
```

2.2 Control / DF Summary

After cleaning the dataset, this chunk creates backup files that can be used later if needed and help preserve progress in case the code or working environment is lost. Furthermore, the dataframes are summarised and checked to provide a better overview of the data and to verify whether the previous cleaning steps were applied correctly.

```r
# ------
cand_clean <- cand
cand_real <- cand_clean |> filter(others == 0)
# --- 1) Overall summary (all rows, incl. Others) ---
summary_all <- cand_clean |>
  summarise(
    rows_total = n(),
    elections_total = n_distinct(election_id),
```

```r
    cantons = n_distinct(canton),
    years = n_distinct(year),
    chambers = n_distinct(chamber),

    others_n = sum(others == 1),
    others_share = mean(others == 1),

    vote_share_missing = mean(is.na(vote_share)),
    total_votes_missing = mean(is.na(total_votes)),
    personal_votes_missing = mean(is.na(personal_votes))
  )

# --- 2) Candidate-only summary (excl. Others) ---
summary_candidates <- cand_real |>
  summarise(
    candidates_n = n(),
    elections_candidates = n_distinct(election_id),

    # Gender coverage and distribution
    gender_missing_share = mean(is.na(gender)),
    male_share = mean(str_to_lower(gender) == "male", na.rm = TRUE),
    female_share = mean(str_to_lower(gender) == "female", na.rm = TRUE),

    # Age coverage
    age_missing_share = mean(is.na(age_election)),
    age_mean = mean(age_election, na.rm = TRUE),
    age_median = median(age_election, na.rm = TRUE),
    age_min = suppressWarnings(min(age_election, na.rm = TRUE)),
    age_max = suppressWarnings(max(age_election, na.rm = TRUE)),

    # RIS coverage
    ris_available_share = mean(!is.na(ris)),
    ris_missing_share = mean(is.na(ris)),

    # Smartvote availability (if coded 0/1)
    smartvote_available_share = mean(smartvote %in% c(0L, 1L), na.rm = TRUE),
    smartvote_yes_share = mean(smartvote == 1L, na.rm = TRUE)
  )

# --- 3) Coverage by election (excl. Others) ---
coverage_by_election <- cand_real |>
  group_by(election_id, canton, year, chamber) |>
```

```r
  summarise(
    n_candidates = n(),
    ris_share = mean(!is.na(ris)),
    gender_share = mean(!is.na(gender)),
    age_share = mean(!is.na(age_election)),
    vote_share = mean(!is.na(vote_share)),
    .groups = "drop"
  ) |>
  arrange(year, canton, chamber)

# --- 4) Coverage by canton-year-chamber (excl. Others) ---
coverage_by_group <- cand_real |>
  group_by(canton, chamber, year) |>
  summarise(
    n_candidates = n(),
    ris_share = mean(!is.na(ris)),
    gender_share = mean(!is.na(gender)),
    age_share = mean(!is.na(age_election)),
    .groups = "drop"
  ) |>
  arrange(canton, chamber, year)

# --- 5) Quick "worst elections" diagnostics (excl. Others) ---
worst_ris <- coverage_by_election |>
  arrange(ris_share) |>
  slice_head(n = 15)

worst_age <- coverage_by_election |>
  arrange(age_share) |>
  slice_head(n = 15)

# --- 6) Print summaries to the report (nice console output) ---
cat("\n--- SUMMARY (ALL ROWS, incl. Others) ---\n")
```

```
--- SUMMARY (ALL ROWS, incl. Others) ---
```

```r
print(summary_all)
```

```
# A tibble: 1 x 10
  rows_total elections_total cantons years chambers others_n others_share
```

```
       <int>             <int>    <int> <int>     <int>     <int>        <dbl>
1        212                72        8     6         2        34        0.160
# i 3 more variables: vote_share_missing <dbl>, total_votes_missing <dbl>,
#   personal_votes_missing <dbl>
```

```r
cat("\n--- SUMMARY (CANDIDATES ONLY, excl. Others) ---\n")
```

```
--- SUMMARY (CANDIDATES ONLY, excl. Others) ---
```

```r
print(summary_candidates)
```

```
# A tibble: 1 x 14
  candidates_n elections_candidates gender_missing_share male_share female_share
         <int>                <int>                <dbl>      <dbl>        <dbl>
1          178                   72                    0      0.803        0.197
# i 9 more variables: age_missing_share <dbl>, age_mean <dbl>,
#   age_median <dbl>, age_min <dbl>, age_max <dbl>, ris_available_share <dbl>,
#   ris_missing_share <dbl>, smartvote_available_share <dbl>,
#   smartvote_yes_share <dbl>
```

```r
cat("\n--- WORST RIS COVERAGE (CANDIDATES ONLY) ---\n")
```

```
--- WORST RIS COVERAGE (CANDIDATES ONLY) ---
```

```r
print(worst_ris)
```

```
# A tibble: 15 x 9
   election_id          canton  year chamber n_candidates ris_share gender_share
   <chr>                <chr>  <int> <chr>          <int>     <dbl>        <dbl>
 1 Appenzell Ausserrho~ Appen~  2003 NR                 3         0            1
 2 Appenzell Ausserrho~ Appen~  2003 SR                 1         0            1
 3 Appenzell Innerrhod~ Appen~  2003 NR                 3         0            1
 4 Appenzell Innerrhod~ Appen~  2003 SR                 1         0            1
 5 Basel-Landschaft_20~ Basel~  2003 SR                 2         0            1
 6 Basel-Stadt_2003_SR  Basel~  2003 SR                 7         0            1
 7 Glarus_2003_NR       Glarus  2003 NR                 3         0            1
 8 Nidwalden_2003_NR    Nidwa~  2003 NR                 2         0            1
```

```
 9 Nidwalden_2003_SR    Nidwa~  2003 SR                2          0             1
10 Obwalden_2003_NR     Obwal~  2003 NR                2          0             1
11 Obwalden_2003_SR     Obwal~  2003 SR                1          0             1
12 Uri_2003_NR          Uri     2003 NR                3          0             1
13 Appenzell Innerrhod~ Appen~  2007 NR                1          0             1
14 Appenzell Innerrhod~ Appen~  2007 SR                3          0             1
15 Glarus_2007_NR       Glarus  2007 NR                3          0             1
# i 2 more variables: age_share <dbl>, vote_share <dbl>
```

```r
cat("\n--- WORST AGE COVERAGE (CANDIDATES ONLY) ---\n")
```

```
--- WORST AGE COVERAGE (CANDIDATES ONLY) ---
```

```r
print(worst_age)
```

```
# A tibble: 15 x 9
   election_id          canton  year chamber n_candidates ris_share gender_share
   <chr>                <chr>  <int> <chr>          <int>     <dbl>        <dbl>
 1 Basel-Landschaft_20~ Basel~  2003 SR                2        0             1
 2 Appenzell Ausserrho~ Appen~  2003 NR                3        0             1
 3 Appenzell Ausserrho~ Appen~  2003 SR                1        0             1
 4 Appenzell Innerrhod~ Appen~  2003 NR                3        0             1
 5 Appenzell Innerrhod~ Appen~  2003 SR                1        0             1
 6 Basel-Stadt_2003_SR  Basel~  2003 SR                7        0             1
 7 Glarus_2003_NR       Glarus  2003 NR                3        0             1
 8 Nidwalden_2003_NR    Nidwa~  2003 NR                2        0             1
 9 Nidwalden_2003_SR    Nidwa~  2003 SR                2        0             1
10 Obwalden_2003_NR     Obwal~  2003 NR                2        0             1
11 Obwalden_2003_SR     Obwal~  2003 SR                1        0             1
12 Uri_2003_NR          Uri     2003 NR                3        0             1
13 Appenzell Ausserrho~ Appen~  2007 NR                5        0.2           1
14 Appenzell Ausserrho~ Appen~  2007 SR                2        0.5           1
15 Appenzell Innerrhod~ Appen~  2007 NR                1        0             1
# i 2 more variables: age_share <dbl>, vote_share <dbl>
```

```r
# --- Export cleaned data ---
saveRDS(cand_clean, here("data", "clean", "cand_clean.rds"))
write_csv(cand_clean, here("data", "clean", "cand_clean.csv"))

# Candidate-only version
```

```r
saveRDS(cand_real, here("data", "clean", "cand_candidates_only.rds"))
write_csv(cand_real, here("data", "clean", "cand_candidates_only.csv"))
```

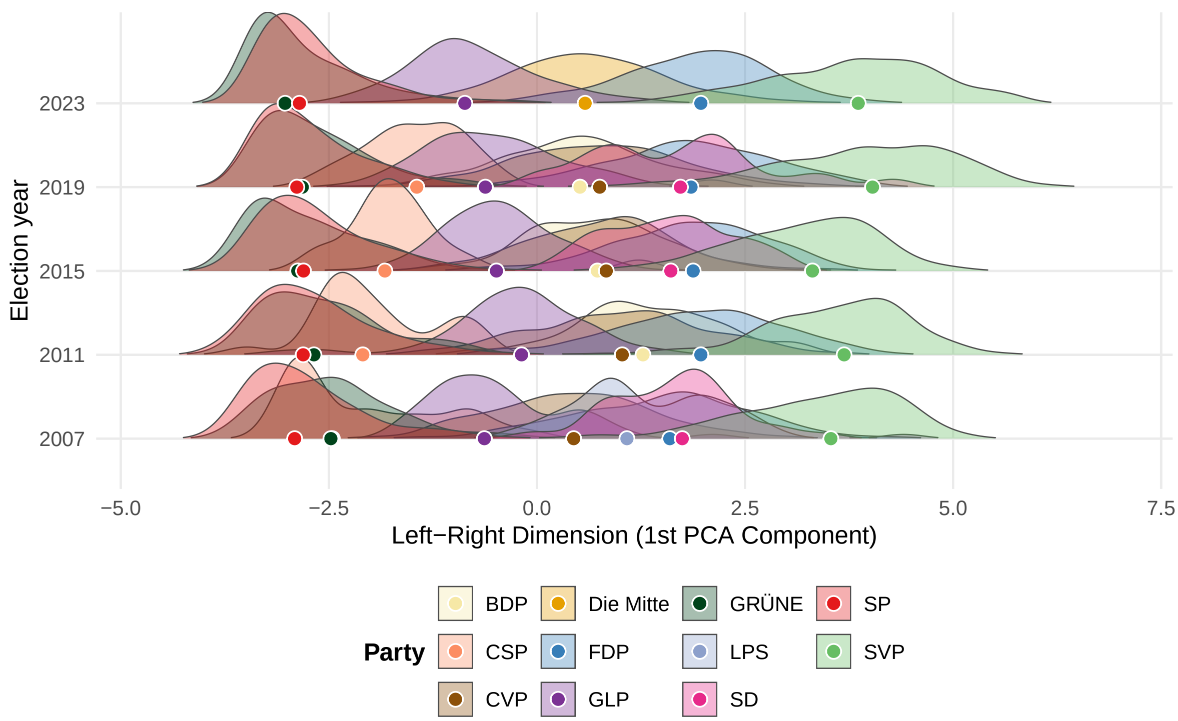
3 RIS Distribution Visualization

This chunk visualizes the distribution of RIS values by election year and party. First, the RIS variable is cleaned by converting non-numeric entries and formatting inconsistencies into usable numeric values. The data is then grouped by election year and party to calculate party-specific median and mean RIS positions. The final plot shows the yearly RIS distributions as density ridges, with points indicating the median RIS position of each party in each election year. Party colors are assigned manually to approximate common Swiss party color schemes while maintaining visual clarity.

```r
base_party_medians <- readr::read_csv2(here("data", "raw",
                      "SV_20XX_FULL_RIS_Input Parteiorientierung.csv"))

base_party_medians <- base_party_medians |>
  mutate(
    Ideology_PC = na_if(Ideology_PC, "#NV"),
    Ideology_PC = na_if(Ideology_PC, "#DIV/0!"),
    Ideology_PC = na_if(Ideology_PC, ""),
    Ideology_PC = str_replace_all(Ideology_PC, ",", "."),
    Ideology_PC = parse_number(Ideology_PC)
  )


party_colors <- c(
  "SP"         = "#E41A1C",
  "SVP"        = "#66BD63",
  "FDP"        = "#377EB8",
  "Die Mitte" = "#E69F00",
  "CVP"        = "#8C510A",
  "BDP"        = "#F6E8A6",
  "GLP"        = "#7B3294",
  "GRÜNE"      = "#00441B",
  "CSP"        = "#FC8D62",
  "LPS"        = "#8DA0CB",
  "SD"         = "#E7298A"
)

#I attempted to apply the official color schemes commonly used in Swiss
```

```
#political analysis, but deviated from them for some parties to ensure
#sufficient visual clarity.

party_year_medians <- base_party_medians |>
  filter(!is.na(Ideology_PC)) |>
  group_by(year, Party_konsolidiert) |>
  summarise(
    party_year_median_ris = median(Ideology_PC, na.rm = TRUE),
    party_year_mean_ris   = mean(Ideology_PC, na.rm = TRUE),
    n_obs                 = n(),
    .groups = "drop"
  )


ParteiPC <- ggplot(base_party_medians, aes(
  x = Ideology_PC,
  y = factor(year),
  fill = Party_konsolidiert
)) +
  geom_density_ridges(
    alpha = 0.35,
    scale = 1.1,
    color = "grey30",
    linewidth = 0.3,
    rel_min_height = 0.01
  ) +
  geom_point(
    data = party_year_medians,
    aes(
      x = party_year_median_ris,   # <-- FIX
      y = factor(year),
      fill = Party_konsolidiert
    ),
    shape = 21,
    size = 3,
    color = "white",
    stroke = 0.6
  ) +
  scale_fill_manual(values = party_colors) +
  labs(
    x = "Left-Right Dimension (1st PCA Component)",
    y = "Election year",
```

```r
    fill = "Party"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "bottom",
    legend.box = "vertical",
    legend.title = element_text(face = "bold"),
    panel.grid.minor = element_blank()
  )

ParteiPC
```



Save the data and plot for further analysis.

```r
ggsave(
  here("output", "figures", "distribution_party_median_by_election.pdf"),
  ParteiPC,
  width = 8,
  height = 5
)
```

```r
ggsave(
  here("output", "figures", "distribution_party_median_by_election.png"),
  ParteiPC,
  width = 8,
  height = 5,
  dpi = 300
)


readr::write_csv(
  party_year_medians,
  here("data", "clean", "party_year_median_ris.csv")
)

saveRDS(
  party_year_medians,
  here("data", "clean", "party_year_median_ris.rds")
)


Test_medians <-
  readr::read_csv(here("data", "clean", "party_year_median_ris.csv"))
```

Party labels were harmonized before calculating the yearly RIS values. Youth parties were pooled with their corresponding main parties in, for example JUSO with SP and JSVP with SVP. This increases the number of observations and reflects the close connection between these organisations.

This step also helps account for changes in smartvote's party coverage over time. Older datasets mainly include larger parties, while newer datasets cover a broader range of parties and youth organisations. Consolidating related parties therefore makes the RIS comparisons across years more consistent.

3.1 Median vs average RIS, Dumbbell plot (Black and White)

Additional plots compare the median and mean RIS values of political parties to assess how strongly the two measures differ from one another.

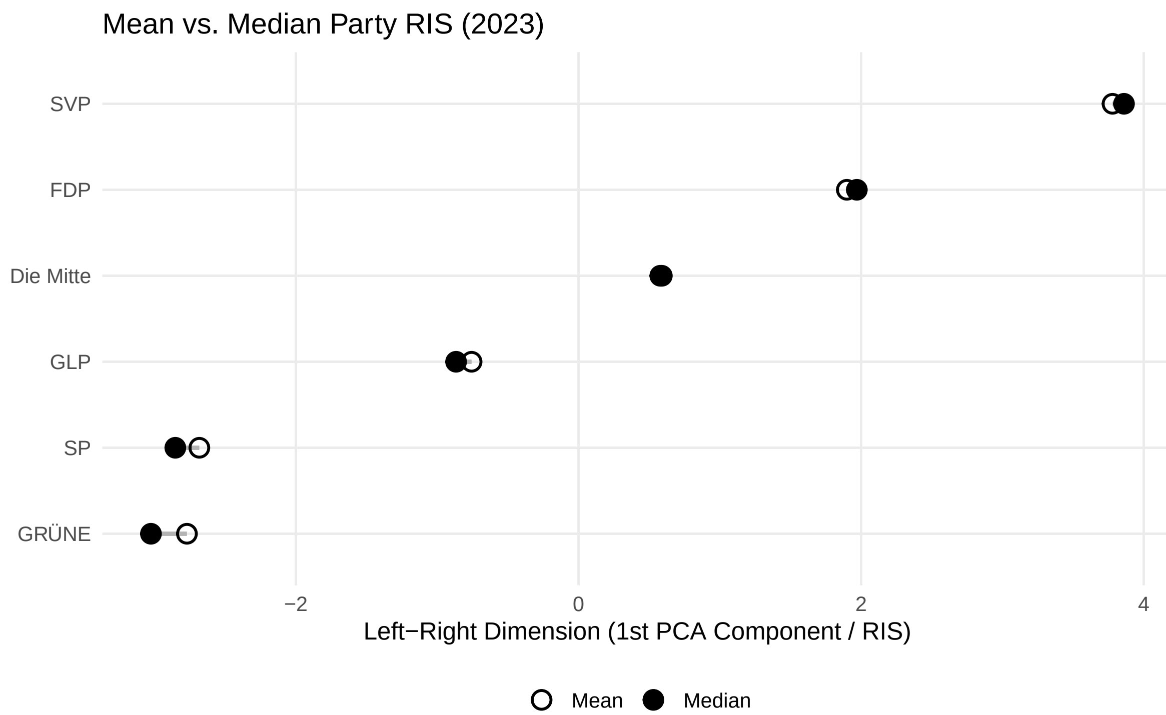
```r
party_stats_2023 <- base_party_medians |>
  filter(year == 2023) |>
  filter(!is.na(Ideology_PC)) |>
```

```r
  group_by(Party_konsolidiert) |>
  summarise(
    median_ris = median(Ideology_PC, na.rm = TRUE),
    mean_ris   = mean(Ideology_PC, na.rm = TRUE),
    n          = n(),
    .groups = "drop"
  ) |>
  # Keep parties with atleast 5 observations
  filter(n >= 5) |>
  arrange(median_ris) |>
  mutate(Party_konsolidiert = factor(Party_konsolidiert,
                                     levels = Party_konsolidiert))

party_stats_2023
```

```
# A tibble: 6 x 4
  Party_konsolidiert median_ris mean_ris     n
  <fct>                   <dbl>    <dbl> <int>
1 GRÜNE                  -3.03    -2.77    610
2 SP                     -2.85    -2.68    671
3 GLP                    -0.866   -0.757   778
4 Die Mitte               0.579    0.590   951
5 FDP                     1.97     1.90    600
6 SVP                     3.86     3.78    502
```

```r
party_stats_2023_long <- party_stats_2023 |>
  select(Party_konsolidiert, median_ris, mean_ris) |>
  pivot_longer(
    cols = c(median_ris, mean_ris),
    names_to = "stat",
    values_to = "value"
  ) |>
  mutate(
    stat = recode(stat,
                  median_ris = "Median",
                  mean_ris   = "Mean")
  )

p_dumbbell_2023 <- ggplot(party_stats_2023) +
  # connector line
  geom_segment(
```

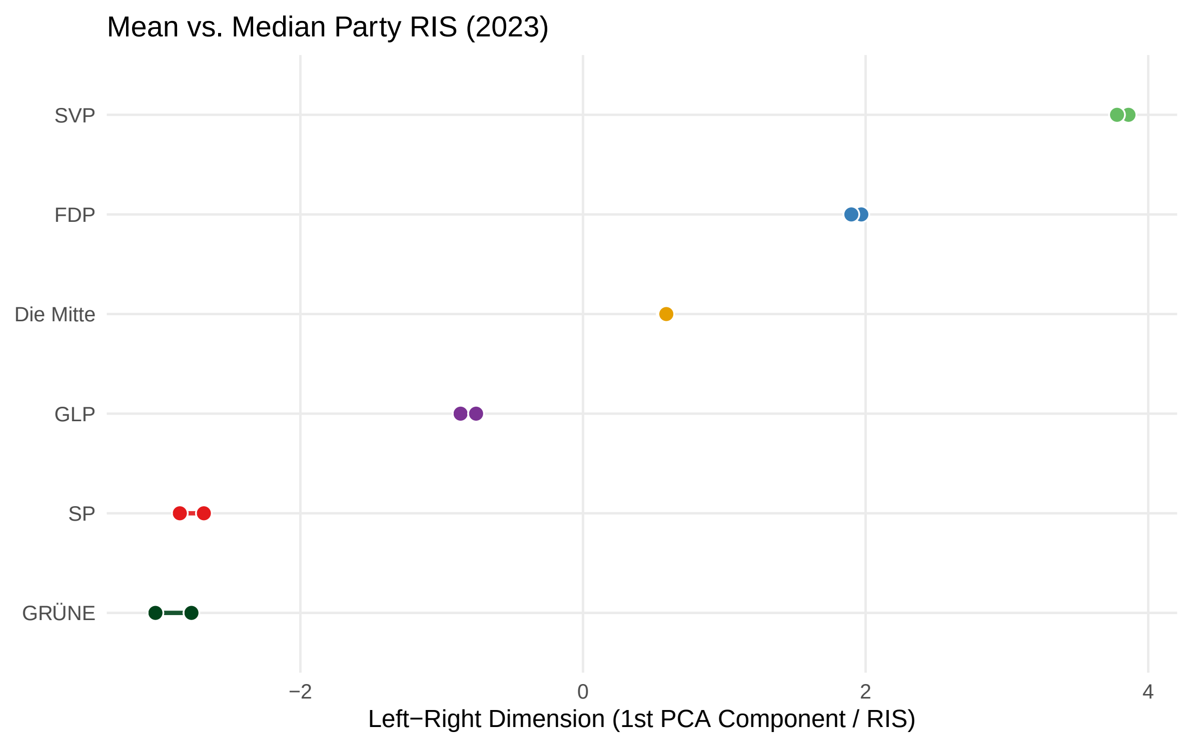
```r
    aes(x = median_ris, xend = mean_ris,
        y = Party_konsolidiert, yend = Party_konsolidiert),
    linewidth = 1, color = "grey70"
  ) +
  # points: mean vs median (clean & readable)
  geom_point(
    data = party_stats_2023_long,
    aes(x = value, y = Party_konsolidiert, shape = stat),
    size = 3.6, stroke = 1.0, color = "black", fill = "white"
  ) +
  scale_shape_manual(
    values = c("Median" = 19, "Mean" = 1),   # filled vs hollow
    name = NULL
  ) +
  labs(
    title = "Mean vs. Median Party RIS (2023)",
    x = "Left-Right Dimension (1st PCA Component / RIS)",
    y = NULL
  ) +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )

p_dumbbell_2023
```



3.2 Median vs average RIS, Dumbbell plot (Party-colors)

```
p_dumbbell_2023_color <- ggplot(party_stats_2023) +
  geom_segment(aes(x = median_ris, xend = mean_ris,
                   y = Party_konsolidiert, yend = Party_konsolidiert,
                   color = Party_konsolidiert),
               linewidth = 1, alpha = 0.9) +
  geom_point(aes(x = median_ris, y = Party_konsolidiert,
                 fill = Party_konsolidiert),
             size = 3.2, shape = 21, color = "white", stroke = 0.6) +
  geom_point(aes(x = mean_ris, y = Party_konsolidiert,
                 fill = Party_konsolidiert),
             size = 3.2, shape = 21, color = "white", stroke = 0.6) +
  scale_color_manual(values = party_colors, guide = "none") +
  scale_fill_manual(values = party_colors, guide = "none") +
  labs(
    title = "Mean vs. Median Party RIS (2023)",
    x = "Left-Right Dimension (1st PCA Component / RIS)",
    y = NULL
```

```
  ) +
  theme_minimal(base_size = 12) +
  theme(panel.grid.minor = element_blank())

p_dumbbell_2023_color
```



In the 2023 plot, the mean and median values are very close for most parties, suggesting that the party-level RIS positions are not strongly affected by extreme values or skewed distributions.

```
ggsave(
  here("output", "figures", "p_dumbbell_2023_color.pdf"),
  p_dumbbell_2023_color,
  width = 8,
  height = 5
)

ggsave(
  here("output", "figures", "p_dumbbell_2023_color.png"),
  p_dumbbell_2023_color,
  width = 8,
```

```r
  height = 5,
  dpi = 300
)
ggsave(
  here("output", "figures", "p_dumbbell_2023.pdf"),
  p_dumbbell_2023,
  width = 8,
  height = 5
)

ggsave(
  here("output", "figures", "p_dumbbell_2023.png"),
  p_dumbbell_2023,
  width = 8,
  height = 5,
  dpi = 300
)
```

4 Regimes

After deciding to split the analysis into three parts, uncontested elections, two-candidate elections, and multi-candidate elections, this chunk creates the corresponding election regimes. These regime variables are then used in the following analysis steps to separate elections by the number of candidates and apply the appropriate models and comparisons.

```r
cand_real <- cand |>
  filter(others == 0)

election_n <- cand_real |>
  count(election_id, name = "n_candidates")

cand_real <- cand_real |>
  left_join(election_n, by = "election_id") |>
  mutate(
    regime = case_when(
      n_candidates == 1 ~ "One-candidate",
      n_candidates == 2 ~ "Two-candidate",
      n_candidates >= 3 ~ "Multi-candidate"
    )
  )
```

4.1 Analysis Dataframe

This chunk saves the final analysis-ready base dataset, cand_analysis. It contains the relevant candidate, election, outcome, RIS, and regime variables and serves as the main dataset for all subsequent filtering and analysis steps.

```
# Final analysis-ready dataset with regimes
cand_analysis <- cand_real |>
  select(
    name,
    id_ba,
    election_id,
    canton,
    year,
    chamber,
    party,
    pol_party_value,
    gender,
    age_election,
    win,
    re_election,
    vote_share,
    ris,
    n_candidates,
    regime,
    smartvote
  )
```

This chunk creates the binary control variables used in the regressions: female, party_affiliated, and re_election_bin. The updated dataset is saved as both an RDS and CSV file, and simple cross-tables are used to verify that the recoding worked correctly.

```
cand_analysis <- cand_analysis |>
  mutate(
    female = case_when(
      gender == "Female" ~ 1,
      gender == "Male"   ~ 0,
      TRUE               ~ NA_real_
    )
  )

cand_analysis <- cand_analysis |>
```

```r
  mutate(
    party_affiliated = case_when(
      party == "Independent" ~ 0,
      !is.na(party)          ~ 1,
      TRUE                   ~ NA_real_
    )
  )

cand_analysis <- cand_analysis |>
  mutate(
    re_election = str_trim(re_election),
    re_election_bin = case_when(
      re_election == "Yes" ~ 1,
      re_election == "No"  ~ 0,
      TRUE                 ~ NA_real_
    )
  )
# Save as RDS (best for R, keeps types)
saveRDS(
  cand_analysis,
  here("data", "clean", "cand_analysis.rds")
)

# Save as CSV (for transparency / appendix / backup)
write_csv(
  cand_analysis,
  here("data", "clean", "cand_analysis.csv")
)

# Check if bins. have been compiled correctly
table(cand_analysis$gender, cand_analysis$female, useNA = "ifany")
```

```
          0   1
  Female  0  35
  Male  143   0
```

```r
table(cand_analysis$party, cand_analysis$party_affiliated, useNA = "ifany")
```

```
                                  0   1
```

```
Aufrecht                         0  1
BDP                              0  3
CSP                              0  2
CVP                              0 37
FDP                              0 35
GLP                              0  2
GRÜNE                            0  7
Grünes Appenzellerland (Gral)    0  1
Independent                     15  0
JUSO                             0  2
LDP                              0  1
LPS                              0  2
SD                               0  3
SP                               0 26
SVP                              0 36
VA                               0  5
```

```r
table(cand_analysis$re_election, cand_analysis$re_election_bin, useNA = "ifany")
```

```
        0   1 <NA>
No     26   0    0
Yes     0  46    0
<NA>    0   0  106
```

This code applies the main sample restrictions for the analysis. Appenzell Innerrhoden is excluded because its Landsgemeinde elections are based on open assembly voting rather than anonymous ballots, making them institutionally different from the other elections in the sample. In addition, elections before 2007 are removed because smartvote candidate data are not consistently available before this year, preventing the construction of comparable Relative Ideology Scores (RIS). (The category "Others" has already been filtered.)

```r
cand_analysis_clean <- cand_analysis |>
  filter(!str_detect(canton, "Innerrhoden")) |>
    filter(year >= 2007) |>
  mutate(ris = if_else(ris == 0, NA_real_, ris))
```

After parsing the data, I identified a small number of wrongly assigned or missing party median values. To correct this, I created party_median_corrections, a dataframe containing the yearly party medians calculated from the PCA-based RIS values. These corrected medians are then joined to the analysis dataset by year and party and used to update pol_party_value.

In addition, the party label "CVP" is recoded to "Die Mitte" for 2023 to reflect the party name used in that election year. This ensures that party-level ideological values are assigned consistently and do not create problems in the later analysis.

```
party_median_corrections <- tribble(
  ~year, ~party, ~median_pc,
  2023, "SVP",        3.860102392,
  2023, "SP",        -2.853187797,
  2023, "GRÜNE",     -3.025733348,
  2023, "GLP",       -0.866285937,
  2023, "FDP",        1.969615109,
  2023, "Die Mitte",  0.578698095,
  2019, "SVP",        4.031765106,
  2019, "SP",        -2.885054321,
  2019, "SD",         1.724332073,
  2019, "GRÜNE",     -2.819653349,
  2019, "GLP",       -0.619653097,
  2019, "FDP",        1.851145669,
  2019, "CVP",        0.753787684,
  2019, "CSP",       -1.443629111,
  2019, "BDP",        0.517464083,
  2015, "SVP",        3.309910114,
  2015, "SP",        -2.804559377,
  2015, "SD",         1.608621009,
  2015, "GRÜNE",     -2.873415605,
  2015, "GLP",       -0.486657328,
  2015, "FDP",        1.876966543,
  2015, "CVP",        0.832736645,
  2015, "CSP",       -1.826875407,
  2015, "BDP",        0.726544249,
  2011, "SVP",        3.690744921,
  2011, "SP",        -2.808903191,
  2011, "GRÜNE",     -2.679462496,
  2011, "GLP",       -0.185025909,
  2011, "FDP",        1.969357118,
  2011, "CVP",        1.024805920,
  2011, "CSP",       -2.092550494,
  2011, "BDP",        1.274032831,
  2007, "SVP",        3.531691254,
  2007, "SP",        -2.910873847,
  2007, "SD",         1.745988610,
  2007, "LPS",        1.081457913,
  2007, "GRÜNE",     -2.475721728,
```

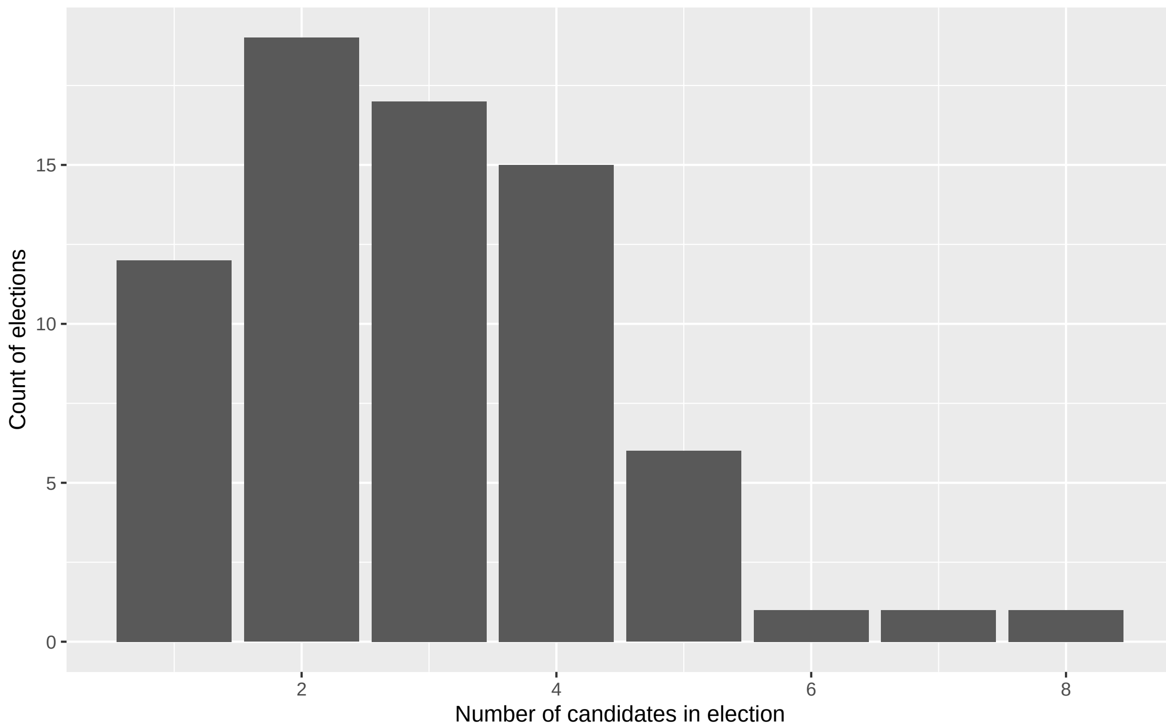
```r
  2007, "GLP",        -0.631705681,
  2007, "FDP",         1.596538279,
  2007, "CVP",         0.441420340,
  2007, "CSP",        -2.458123967
)

cand_analysis_clean <- cand_analysis_clean |>
  mutate(
    party = if_else(
      year == 2023 & party == "CVP",
      "Die Mitte",
      party
    )
  ) |>
  left_join(
    party_median_corrections,
    by = c("year", "party")
  ) |>
  mutate(
    pol_party_value = median_pc
  ) |>
  select(-median_pc)
```

Before filtering, the dataset contained 212 candidate observations. After applying the sample restrictions, the dataset is reduced to 129 observations. Although this is a substantial decrease, keeping the excluded observations would mainly inflate the sample size without improving the quality of the analysis. Since these cases are less comparable or contain less reliable information, including them could negatively affect the statistical validity of the results. Therefore, I consider this restriction a necessary and justified step.

4.2 Visualization pre filtering

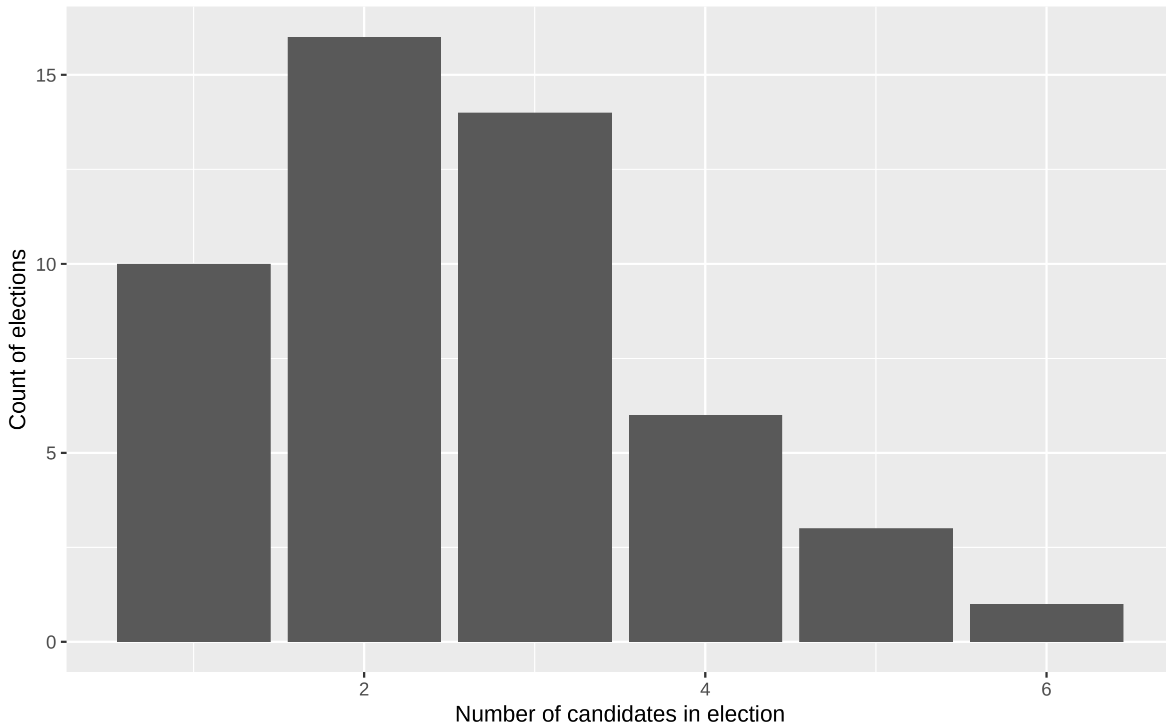
```r
p_candidates_hist_prefilter <- cand_clean |>
  count(election_id, canton, year, chamber, name = "n_candidates") |>
  ggplot(aes(x = n_candidates)) +
  geom_bar() +
  labs(
    x = "Number of candidates in election",
    y = "Count of elections"
  )
```

```
ggsave(
  filename = here("output", "figures", "hist_n_candidates_prefilter.pdf"),
  plot = p_candidates_hist_prefilter,
  width = 8,
  height = 5
)

ggsave(
  filename = here("output", "figures", "hist_n_candidates_prefilter.png"),
  plot = p_candidates_hist_prefilter,
  width = 8,
  height = 5,
  dpi = 300
)
p_candidates_hist_prefilter
```



4.3 Visualization post filtering

```r
p_candidates_hist_postfilter <- cand_analysis_clean |>
  count(election_id, canton, year, chamber, name = "n_candidates") |>
  ggplot(aes(x = n_candidates)) +
  geom_bar() +
  labs(
    x = "Number of candidates in election",
    y = "Count of elections"
  )
ggsave(
  filename = here("output", "figures", "hist_n_candidates_postfilter.pdf"),
  plot = p_candidates_hist_postfilter,
  width = 8,
  height = 5
)

ggsave(
  filename = here("output", "figures", "hist_n_candidates_postfilter.png"),
  plot = p_candidates_hist_postfilter,
  width = 8,
  height = 5,
  dpi = 300
)
p_candidates_hist_postfilter
```



5 Analysis

The following analysis is based on the cleaned candidate-level sample constructed above. The residual category “Others” has already been removed, as it consists of aggregated, non-identifiable candidates without individual-level information such as ideology, gender, age, or incumbency status. These observations cannot be linked to smartvote data and are therefore unsuitable for the candidate-level analysis.

Elections are classified by the number of remaining identifiable candidates: one-candidate contests are treated as uncontested or quasi-uncontested elections, two-candidate contests as binary elections, and contests with three or more candidates as multi-candidate elections. The sample is restricted to elections from 2007 onward because comparable smartvote-based ideological positions are not consistently available before 2007.

```
# Save as RDS (best for R, keeps types)
saveRDS(
  cand_analysis_clean,
  here("data", "clean", "cand_analysis_clean.rds")
)

# Save as CSV (for transparency / appendix / backup)
```

```
write_csv(
  cand_analysis_clean,
  here("data", "clean", "cand_analysis_clean.csv")
)
```

5.1 One Candidate (Uncontested) elections

```
One_candidate_elections <- cand_analysis_clean |>
  filter(regime=="One-candidate")

One_candidate_elections <- One_candidate_elections |>
  mutate(
    silent_type = case_when(
      is.na(vote_share) ~ "Truly uncontested (no vote share recorded)",
      TRUE              ~ "Quasi-silent (competition only via 'Others')"
    ),
    vote_share_pct = if_else(is.na(vote_share), NA_real_, 100 * vote_share),
    ris_available = !is.na(ris)
  )

write_csv(
  One_candidate_elections,
  here("data", "clean", "One_candidate_elections.csv")
)
saveRDS(
  One_candidate_elections,
  here("data", "clean", "One_candidate_elections.rds")
)
```

Appendix table with the most important information about the candidates

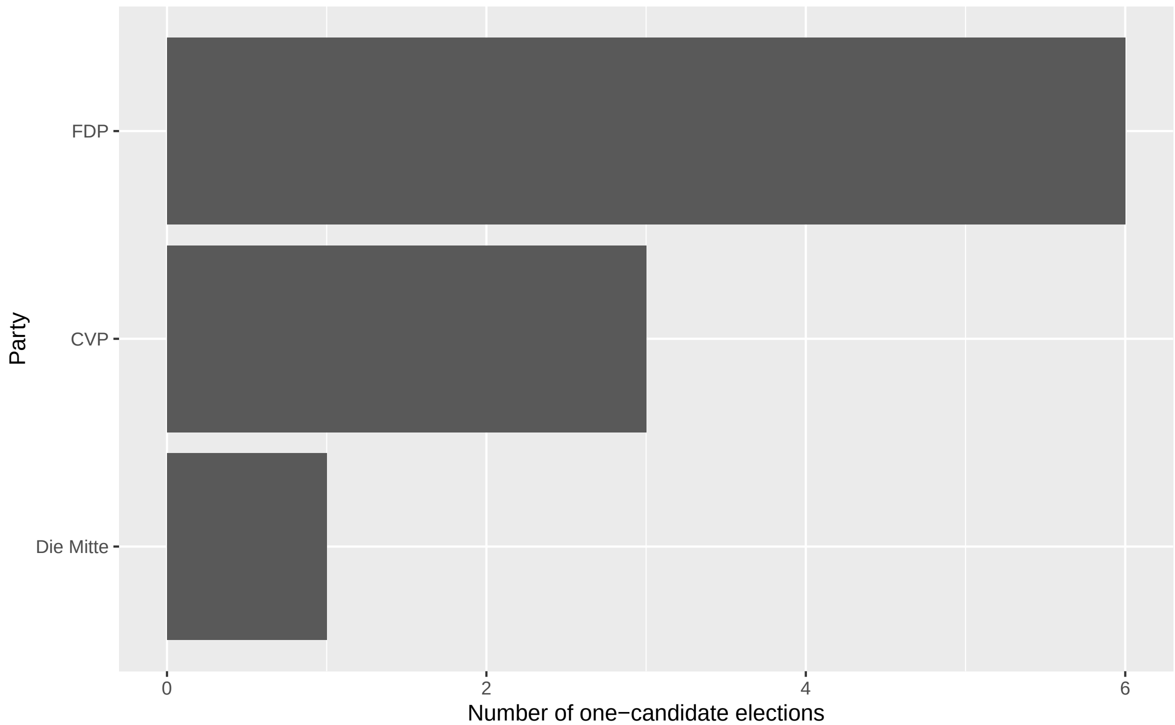
```
tab_election <- One_candidate_elections |>
  transmute(
    election_id,
    canton,
    year,
    chamber,
    gender,
    age_election,
    party,
```

```
    re_election,
    RIS = ris,
    vote_share_pct,
    silent_type
  ) |>
  arrange(year, chamber, canton)

gt_election_oneseat <- tab_election |>
  gt(groupname_col = "year") |>
  cols_label(
    election_id = "Election ID",
    canton = "Canton",
    chamber = "Chamber",
    gender = "Gender",
    age_election = "Age at Election",
    party = "Party",
    re_election = "Incumbent?",
    RIS = "RIS",
    vote_share_pct = "Vote share (%)",
    silent_type = "Silent-election type"
  ) |>
  fmt_number(columns = RIS, decimals = 2) |>
  fmt_number(columns = vote_share_pct, decimals = 1) |>
  sub_missing(columns = c(RIS, vote_share_pct), missing_text = "NA") |>
  tab_header(
    title = "One-candidate elections in the analytical sample",
    subtitle = "Distinguishing truly uncontested from quasi-silent elections"
  )
gtsave(
  gt_election_oneseat,
  filename = here("output", "tables", "one_candidate_table.png"),
  zoom = 2
)
```

One Candidate Party distribution bar plot (Black and white)

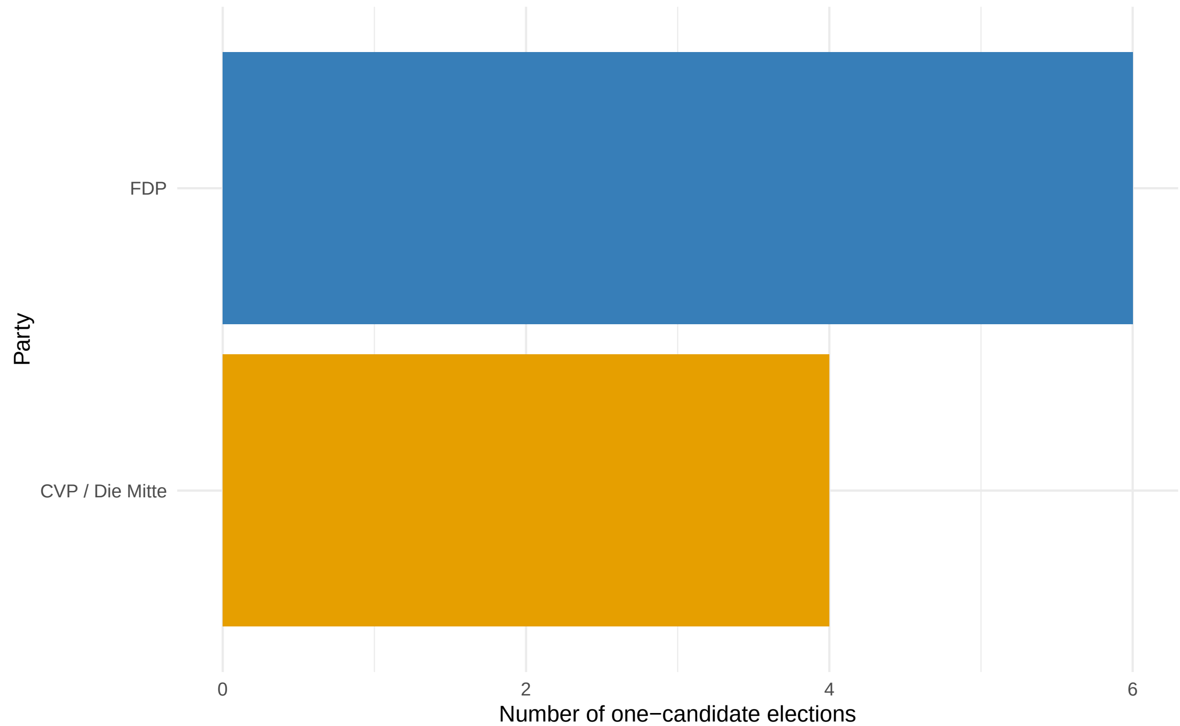
```
p_party <- One_candidate_elections |>
  count(party, name = "n") |>
  arrange(desc(n)) |>
  ggplot(aes(x = reorder(party, n), y = n)) +
  geom_col() +
  coord_flip() +
```

```
  labs(
    x = "Party",
    y = "Number of one-candidate elections",
  )

ggsave(
  filename = here("output", "figures", "one_candidate_party_bar.png"),
  plot = p_party,
  width = 6.5,
  height = 4.2,
  dpi = 300
)

ggsave(
  filename = here("output", "figures", "one_candidate_party_bar.pdf"),
  plot = p_party,
  width = 6.5,
  height = 4.2
)
p_party
```



One Candidate Party distribution bar plot and clean visualization (Colors)

```r
p_party_color <- One_candidate_elections |>
  mutate(
    party_plot = case_when(
      party %in% c("CVP", "Die Mitte") ~ "CVP / Die Mitte",
      TRUE ~ party
    )
  ) |>
  count(party_plot, name = "n") |>
  ggplot(aes(x = reorder(party_plot, n), y = n, fill = party_plot)) +
  geom_col() +
  coord_flip() +
  scale_fill_manual(
    values = c(
      "CVP / Die Mitte" = unname(party_colors["Die Mitte"]),
      "FDP" = unname(party_colors["FDP"])
    )
  ) +
  labs(
    x = "Party",
    y = "Number of one-candidate elections"
  ) +
  theme_minimal() +
  guides(fill = "none")

ggsave(
  filename = here("output", "figures", "one_candidate_party_bar_color.png"),
  plot = p_party_color,
  width = 6.5,
  height = 4.2,
  dpi = 300
)

ggsave(
  filename = here("output", "figures", "one_candidate_party_bar_color.pdf"),
  plot = p_party_color,
  width = 6.5,
  height = 4.2
)

p_party_color
```



5.2 Two-Candidate elections

```
Two_candidate_elections <- cand_analysis_clean |>
  filter(regime=="Two-candidate")

write_csv(
  Two_candidate_elections,
  here("data", "clean", "Two_candidate_elections.csv")
)
saveRDS(
  Two_candidate_elections,
  here("data", "clean", "Two_candidate_elections.rds")
)
```

5.2.1 Two-Candidate analysis

This step constructs ideol_distance_0, the absolute distance of each candidate's RIS from the ideological center at zero. Larger values indicate more ideologically distant candidates, regardless of direction. Where individual RIS values are missing, party-year median scores are used as proxies. Vote share is used as the regression outcome instead of a binary win/loss variable to preserve information on close electoral results.

```
two_candidate_ideology <- Two_candidate_elections |>
  mutate(
    # use observed RIS if available, otherwise the party-year proxy
    ris_used = coalesce(ris, as.numeric(pol_party_value)),

    ideology_source = case_when(
      !is.na(ris) ~ "RIS",
      is.na(ris) & !is.na(pol_party_value) ~ "Party proxy",
      TRUE ~ "Missing"
    ),

    re_election_bin = case_when(
      re_election == "Yes" ~ 1,
      TRUE ~ 0
    ),

    ideol_distance_0 = abs(ris_used)
  ) |>
  filter(!is.na(ris_used))
```

This code estimates four OLS regression models for two-candidate elections, using candidate vote share as the dependent variable. Models (1) and (2) rely only on observed candidate-level RIS values, while Models (3) and (4) additionally use party-year ideology proxies where individual RIS values are missing. The table also adds descriptive rows for the centrist candidate win rate and average winner/loser vote shares. The final regression table is exported as a LaTeX file and can be loaded directly into Overleaf.

```
# --------------------------------------------------
# 1) Regression models two candidate election
# --------------------------------------------------
m1 <- lm(
  vote_share ~ ideol_distance_0,
  data = filter(two_candidate_ideology, ideology_source == "RIS")
)

m2 <- lm(
  vote_share ~ ideol_distance_0 + re_election_bin + female +
    age_election + party_affiliated,
```

```
  data = filter(two_candidate_ideology, ideology_source == "RIS")
)

m3 <- lm(
  vote_share ~ ideol_distance_0,
  data = two_candidate_ideology
)

m4 <- lm(
  vote_share ~ ideol_distance_0 + re_election_bin + female +
    age_election + party_affiliated,
  data = two_candidate_ideology
)

# --------------------------------------------------
# 2) Descriptive statistics for the table
# --------------------------------------------------

# centrist candidate win rate
centrist_winrate <- two_candidate_ideology |>
  group_by(election_id) |>
  summarise(
    winner_distance = ideol_distance_0[win == 1][1],
    loser_distance  = ideol_distance_0[win == 0][1],
    winner_more_centrist = winner_distance < loser_distance,
    .groups = "drop"
  ) |>
  summarise(
    centrist_winrate = mean(winner_more_centrist, na.rm = TRUE)
  ) |>
  pull(centrist_winrate)

# winner/loser average vote shares
vote_share_stats <- two_candidate_ideology |>
  mutate(status = if_else(win == 1, "Winner", "Loser")) |>
  group_by(status) |>
  summarise(
    mean_vote_share = mean(vote_share, na.rm = TRUE),
    .groups = "drop"
  )

winner_mean <- vote_share_stats |>
```

```r
  filter(status == "Winner") |>
  pull(mean_vote_share)

loser_mean <- vote_share_stats |>
  filter(status == "Loser") |>
  pull(mean_vote_share)

# --------------------------------------------------
# 3) Extra rows for the regression table
# --------------------------------------------------

extra_rows_compact <- tribble(
  ~term, ~`(1)`, ~`(2)`, ~`(3)`, ~`(4)`,
  "Centrist candidate wins",
  sprintf("%.0f%%", 100 * centrist_winrate), "", "", "",
  "Mean vote share, winners",
  sprintf("%.3f", winner_mean), "", "", "",
  "Mean vote share, losers",
  sprintf("%.3f", loser_mean), "", "", ""
)

# --------------------------------------------------
# 4) Nice coefficient names
# --------------------------------------------------

coef_names <- c(
  "(Intercept)" = "Constant",
  "ideol_distance_0" = "Ideological distance",
  "re_election_bin" = "Incumbent",
  "female" = "Female",
  "age_election" = "Age",
  "party_affiliated" = "Party affiliated"
)

gof_map_compact <- tribble(
  ~raw, ~clean, ~fmt,
  "nobs", "Observations", 0,
  "r.squared", "R-squared", 3
)

# --------------------------------------------------
# 5) LaTeX table for Overleaf
```

```r
# ---------------------------------------------------

modelsummary(
  list(
    "(1)" = m1,
    "(2)" = m2,
    "(3)" = m3,
    "(4)" = m4
  ),
  coef_map = coef_names,
  statistic = "({std.error})",
  stars = c('*' = .1, '**' = .05, '***' = .01),
  gof_map = gof_map_compact,
  add_rows = extra_rows_compact,
  title = "Two-candidate elections: ideological distance and vote share",
  notes = c(
    "Dependent variable: candidate vote share.",
    "Models (1) and (2) include only candidates with observed RIS values.",
    "Models (3) and (4) use party-year ideology proxies where RIS is unavailable."
  ),
  escape = FALSE,
  output =
    here("output", "tables", "tab_two_candidate_regressions_compact.tex")
)
```

Additional clarifications regarding the table caption and explanatory notes were added directly to the LaTeX table output in Overleaf.

5.2.2 Two-Candidate visualization

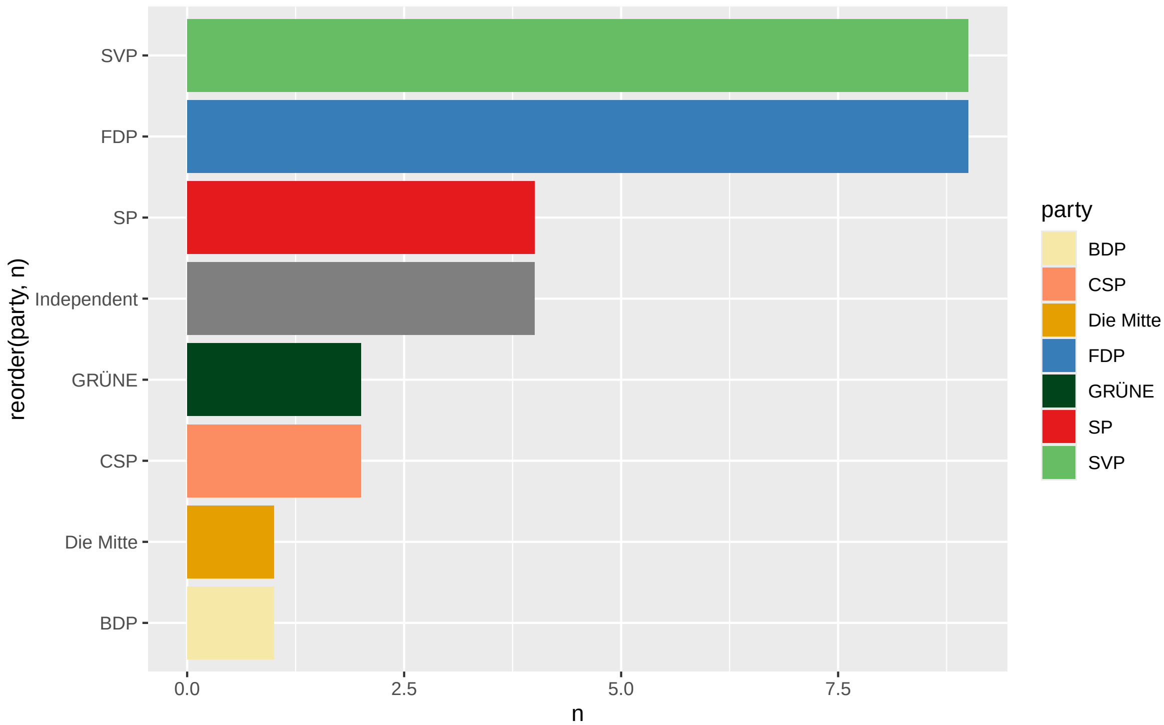
This plot was created as an exploratory visualization to inspect the party distribution among candidates in two-candidate elections. It is not included in the final thesis output, but was used during the analysis process to better understand the composition of the sample.

```r
p_party_color_two_candidate <- Two_candidate_elections |>
  count(party, name = "n") |>
  arrange(desc(n)) |>
  ggplot(aes(x = reorder(party, n), y = n, fill = party)) +
  geom_col() +
  coord_flip() +
  scale_fill_manual(values = party_colors)
```

```r
  labs(
    x = "Party",
    y = "Number of one-candidate elections"
  ) +
  theme_minimal() +
  guides(fill = "none")
```

```
NULL
```

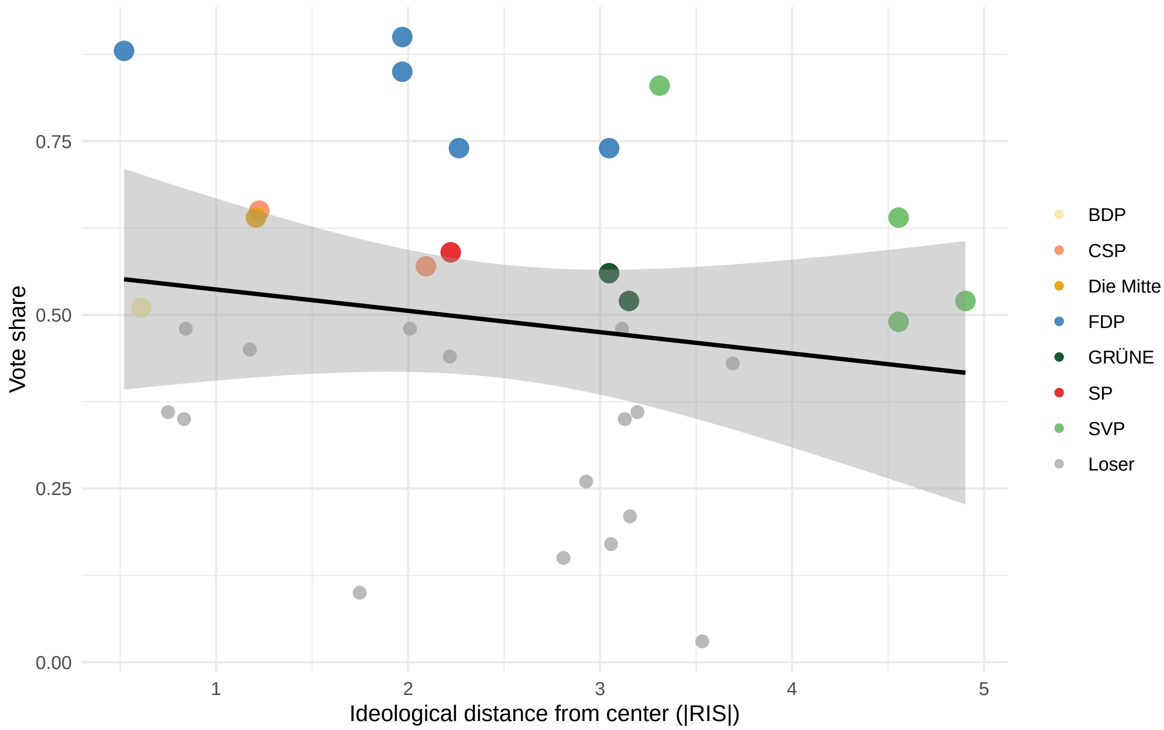
```r
  ggsave(
  filename = here("output", "figures", "Two_candidate_party_bar_color.png"),
  plot = p_party_color_two_candidate,
  width = 6.5,
  height = 4.2,
  dpi = 300
)

ggsave(
  filename = here("output", "figures", "Two_candidate_party_bar_color.pdf"),
  plot = p_party_color_two_candidate,
  width = 6.5,
  height = 4.2
)
p_party_color_two_candidate
```



This plot visualizes the relationship between candidates' ideological distance from the RIS center and their vote share in two-candidate elections. Winners are highlighted by party color, while losing candidates are shown in grey. The fitted regression line provides a first visual impression of whether more centrist candidates tend to receive higher vote shares.

```
two_candidate_plot <- two_candidate_ideology |>
  mutate(
    plot_colour = if_else(win == 1, party, "Loser")
  )
plot_colors <- c(party_colors, "Loser" = "grey70")
## PLOT with Party colors as the colors for won ellections
PlotTwocandidate_Voteshare_RIS <-
two_candidate_plot |>
  mutate(
    plot_colour = factor(plot_colour,
                         levels = c(
                           "BDP","CSP","Die Mitte",
                           "FDP","GRÜNE","SP","SVP","Loser"
                         ))
  ) |>
  ggplot(aes(x = ideol_distance_0, y = vote_share)) +
```

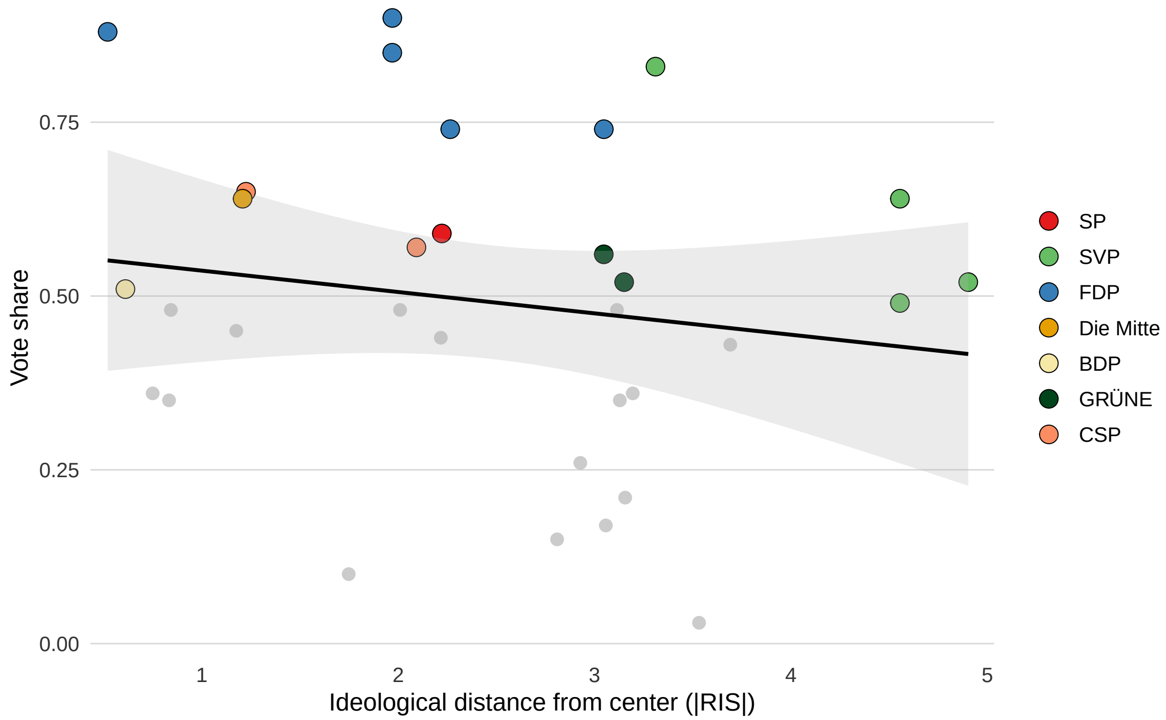
```
  geom_point(aes(color = plot_colour, size = factor(win)), alpha = 0.9) +
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  scale_color_manual(values = plot_colors) +
  scale_size_manual(values = c("0" = 2.5, "1" = 4), guide = "none") +
  labs(
    x = "Ideological distance from center (|RIS|)",
    y = "Vote share",
    color = NULL
  ) +
  theme_minimal()
ggsave(
  filename = here("output", "figures", "Twocandidate_voteshare_RIS.png"),
  plot = PlotTwocandidate_Voteshare_RIS,
  width = 6.5,
  height = 4.2,
  dpi = 300
)

ggsave(
  filename = here("output", "figures", "Twocandidate_voteshare_RIS.pdf"),
  plot = PlotTwocandidate_Voteshare_RIS,
  width = 6.5,
  height = 4.2
)
PlotTwocandidate_Voteshare_RIS
```




```r
PlotTwocandidate_Voteshare_RIS_Improved <- ggplot(
  two_candidate_plot,
  aes(x = ideol_distance_0, y = vote_share)
) +
  geom_point(
    data = \(d) d |> dplyr::filter(win == 0),
    color = "grey75",
    size = 2.4,
    alpha = 0.8
  ) +
  geom_point(
    data = \(d) d |> dplyr::filter(win == 1),
    aes(fill = plot_colour),
    shape = 21,
    color = "black",
    stroke = 0.35,
    size = 4
  ) +
  geom_smooth(
    method = "lm",
    se = TRUE,
```

```
    color = "black",
    linewidth = 0.9,
    fill = "grey70",
    alpha = 0.25
  ) +
  scale_fill_manual(values = plot_colors,
                    breaks = names(plot_colors)
                    [names(plot_colors) != "Loser"]) +
  scale_x_continuous(expand = expansion(mult = c(0.02, 0.03))) +
  scale_y_continuous(
    limits = c(0, NA),
    expand = expansion(mult = c(0.02, 0.03))
  ) +
  labs(
    x = "Ideological distance from center (|RIS|)",
    y = "Vote share",
    fill = NULL,
  ) +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "right",
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    panel.grid.major.y = element_line(color = "grey85", linewidth = 0.35),
    axis.text = element_text(color = "grey20")
  )
ggsave(
  filename =
    here("output", "figures", "PlotTwoicandidate_Voteshare_RIS_Improved.png"),
  plot = PlotTwocandidate_Voteshare_RIS_Improved,
  width = 6.5,
  height = 4.2,
  dpi = 300
)

ggsave(
  filename =
    here("output", "figures", "PlotTwocandidate_Voteshare_RIS_Improved.pdf"),
  plot = PlotTwocandidate_Voteshare_RIS_Improved,
  width = 6.5,
  height = 4.2
)
```

```
PlotTwocandidate_Voteshare_RIS_Improved
```



5.2.3 Multi-Candidate Elections (Three or more Candidate elections)

```
Multi_candidate_elections <- cand_analysis_clean |>
  filter(regime=="Multi-candidate")

write_csv(
  Multi_candidate_elections,
  here("data", "clean", "Multi_candidate_elections.csv")
)
saveRDS(
  Multi_candidate_elections,
  here("data", "clean", "Multi_candidate_elections.rds")
)
```

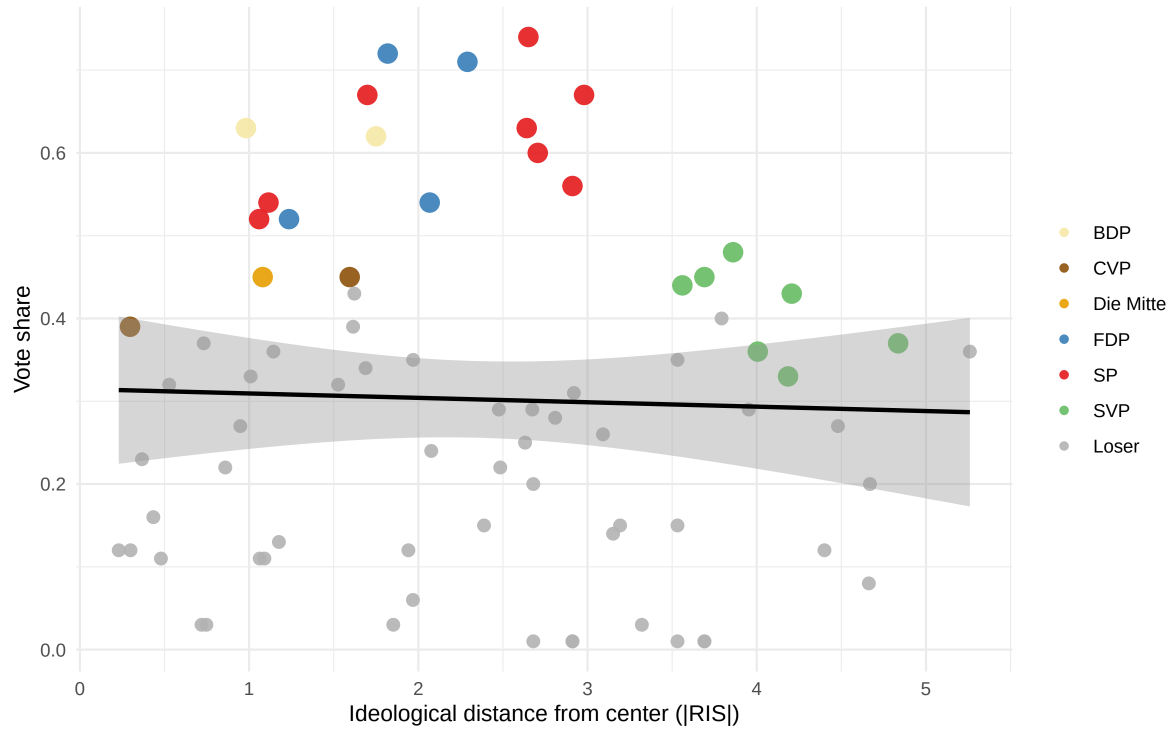
```
Multi_candidate_ideology <- Multi_candidate_elections |>
  mutate(
```

```
    # use observed RIS if available, otherwise the party-year proxy
    ris_used = coalesce(ris, as.numeric(pol_party_value)),

    # flag which source was used
    ideology_source = case_when(
      !is.na(ris) ~ "RIS",
      is.na(ris) & !is.na(pol_party_value) ~ "Party proxy",
      TRUE ~ "Missing"
    ),
# Adjust the relection bin so that all non relected are 0
# not just not reelected winners
      re_election_bin = case_when(
      re_election == "Yes" ~ 1,
      TRUE ~ 0
    ),

    # ideological distance from assumed center / proxy median voter at 0
    ideol_distance_0 = abs(ris_used)
  ) |>
  filter(!is.na(ris_used))
```

This plot visualizes the relationship between candidates' ideological distance from the RIS center and their vote share in multi-candidate elections. Winners are highlighted by party color, while losing candidates are shown in grey. The fitted regression line provides a first visual impression of whether more centrist candidates tend to receive higher vote shares.

```
Multi_candidate_plot <- Multi_candidate_ideology |>
  mutate(
    plot_colour = if_else(win == 1, party, "Loser")
  )
plot_colors <- c(party_colors, "Loser" = "grey70")
## PLOT with Party colors as the colors for won ellections
PlotMulticandidate_Voteshare_RIS <-
Multi_candidate_plot |>
  mutate(
    plot_colour = factor(plot_colour,
                         levels = c(
                           "BDP","CVP","Die Mitte",
                           "FDP","GLP","GRÜNE","CSP",
                           "JUSO","LPS","SP","SVP","Loser"
                         ))
  ) |>
```

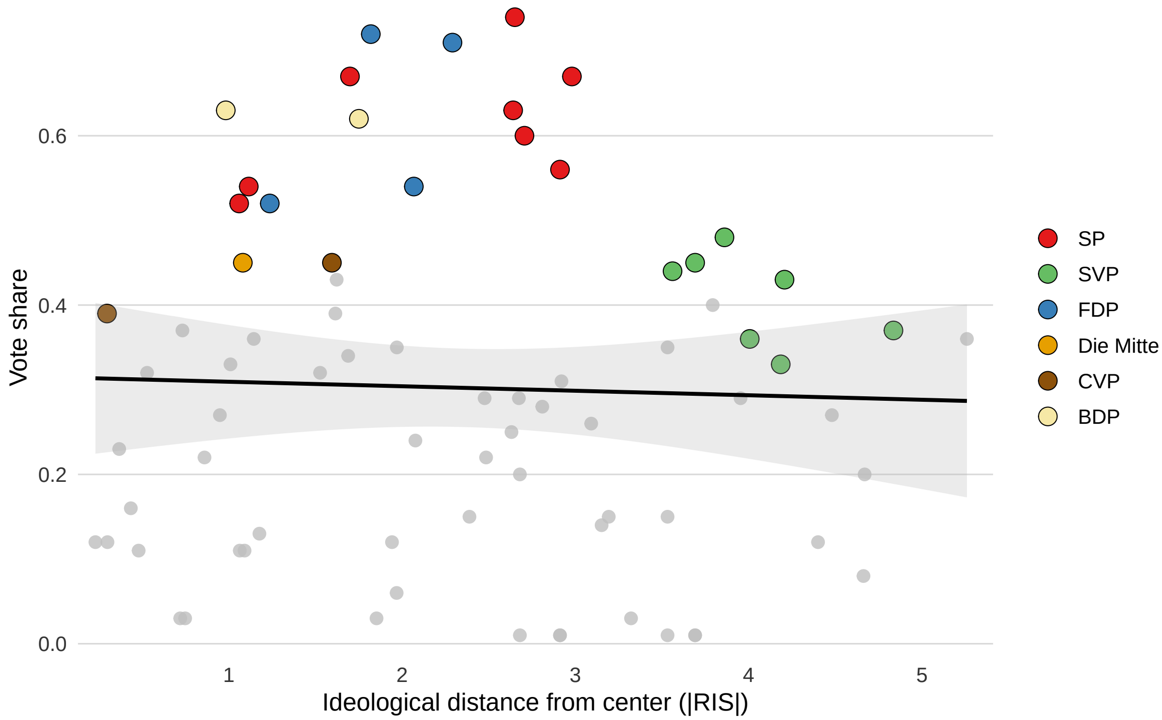
```r
  ggplot(aes(x = ideol_distance_0, y = vote_share)) +
  geom_point(aes(color = plot_colour, size = factor(win)), alpha = 0.9) +
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  scale_color_manual(values = plot_colors) +
  scale_size_manual(values = c("0" = 2.5, "1" = 4), guide = "none") +
  labs(
    x = "Ideological distance from center (|RIS|)",
    y = "Vote share",
    color = NULL
  ) +
  theme_minimal()
ggsave(
  filename = here("output", "figures", "Multicandidate_voteshare_RIS.png"),
  plot = PlotMulticandidate_Voteshare_RIS,
  width = 6.5,
  height = 4.2,
  dpi = 300
)

ggsave(
  filename = here("output", "figures", "Multicandidate_voteshare_RIS.pdf"),
  plot = PlotMulticandidate_Voteshare_RIS,
  width = 6.5,
  height = 4.2
)
PlotMulticandidate_Voteshare_RIS
```




```
PlotMulticandidate_Voteshare_RIS_Improved <- ggplot(
  Multi_candidate_plot,
  aes(x = ideol_distance_0, y = vote_share)
) +
  geom_point(
    data = \(d) d |> dplyr::filter(win == 0),
    color = "grey75",
    size = 2.4,
    alpha = 0.8
  ) +
  geom_point(
    data = \(d) d |> dplyr::filter(win == 1),
    aes(fill = plot_colour),
    shape = 21,
    color = "black",
    stroke = 0.35,
    size = 4
  ) +
  geom_smooth(
    method = "lm",
    se = TRUE,
```

```r
    color = "black",
    linewidth = 0.9,
    fill = "grey70",
    alpha = 0.25
  ) +
  scale_fill_manual(values = plot_colors,
                    breaks = names(plot_colors)
                    [names(plot_colors) != "Loser"]) +
  scale_x_continuous(expand = expansion(mult = c(0.02, 0.03))) +
  scale_y_continuous(
    limits = c(0, NA),
    expand = expansion(mult = c(0.02, 0.03))
  ) +
  labs(
    x = "Ideological distance from center (|RIS|)",
    y = "Vote share",
    fill = NULL,
  ) +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "right",
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    panel.grid.major.y = element_line(color = "grey85", linewidth = 0.35),
    axis.text = element_text(color = "grey20")
  )
ggsave(
  filename =
    here("output", "figures", "PlotMulticandidate_Voteshare_RIS_Improved.png"),
  plot = PlotMulticandidate_Voteshare_RIS_Improved,
  width = 6.5,
  height = 4.2,
  dpi = 300
)

ggsave(
  filename =
    here("output", "figures", "PlotMulticandidate_Voteshare_RIS_Improved.pdf"),
  plot = PlotMulticandidate_Voteshare_RIS_Improved,
  width = 6.5,
  height = 4.2
)
```

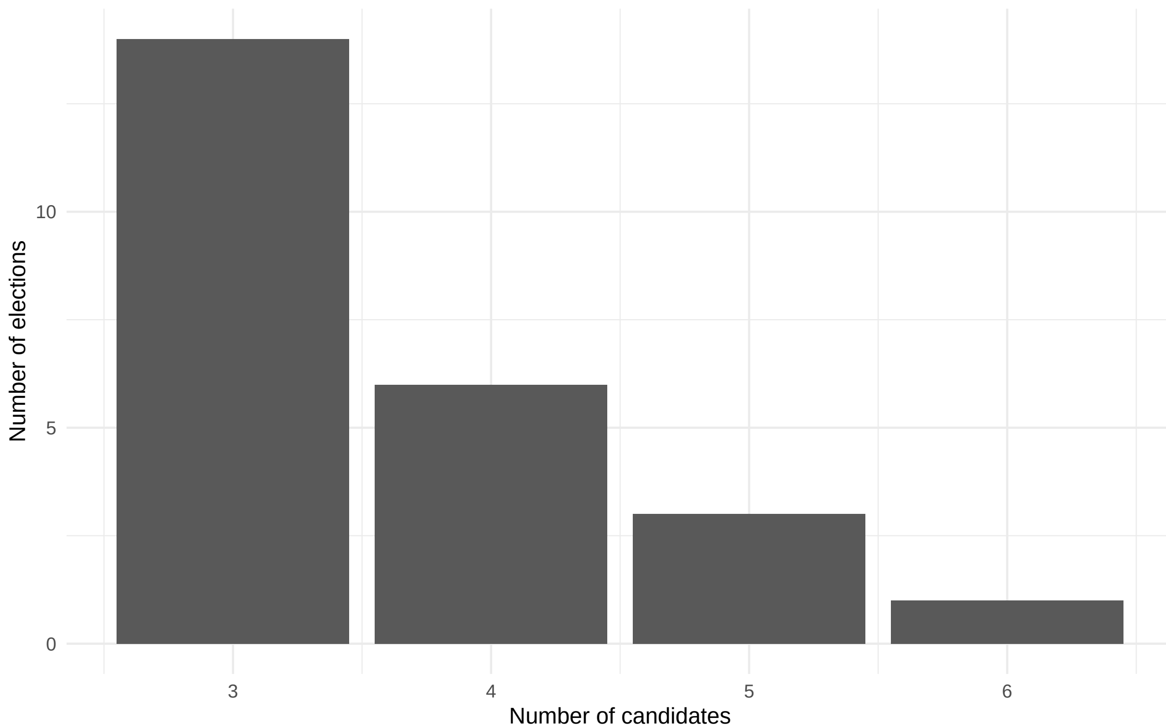
```
PlotMulticandidate_Voteshare_RIS_Improved
```



This plot shows the distribution of election sizes in the multi-candidate sample. It counts how many elections include three or more candidates and provides an overview of how fragmented the candidate fields are in these contests.

```
p_n_candidates <- Multi_candidate_ideology |>
  distinct(election_id, canton, year, chamber, n_candidates) |>
  ggplot(aes(x = n_candidates)) +
  geom_bar() +
  labs(
    x = "Number of candidates",
    y = "Number of elections"
  ) +
  theme_minimal()
ggsave(
  filename = here("output", "figures", "Multi_candidate_n_candidates.png"),
  plot = p_n_candidates,
  width = 6.5,
  height = 4.2,
```

```r
  dpi = 300
)

ggsave(
  filename = here("output", "figures", "Multi_candidate_n_candidates.pdf"),
  plot = p_n_candidates,
  width = 6.5,
  height = 4.2
)

p_n_candidates
```



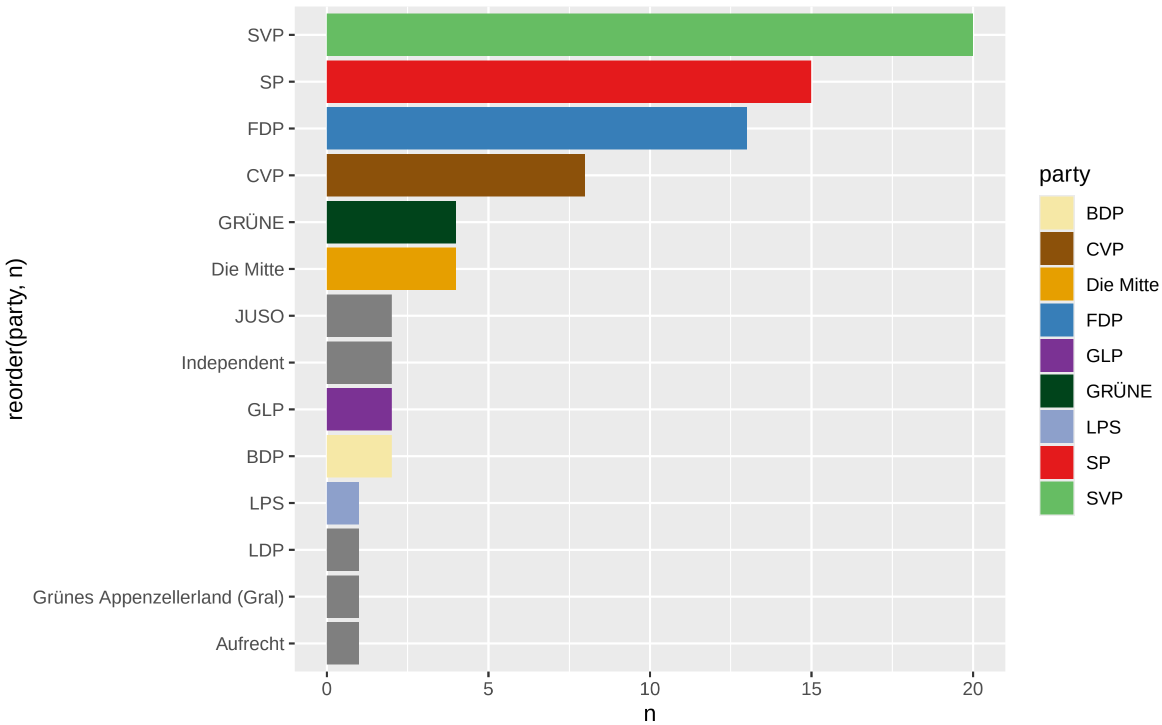
This plot shows the party composition of the multi-candidate election sample. It provides a descriptive overview of which parties appear most frequently in elections with three or more candidates.

```r
p_party_color_multi_candidate <- Multi_candidate_ideology  |>
  count(party, name = "n") |>
  arrange(desc(n)) |>
  ggplot(aes(x = reorder(party, n), y = n, fill = party)) +
```

```
  geom_col() +
  coord_flip() +
  scale_fill_manual(values = party_colors)
  labs(
    x = "Party",
    y = "Number of one-candidate elections"
  ) +
  theme_minimal() +
  guides(fill = "none")
```

```
NULL
```

```
  ggsave(
  filename =
    here("output", "figures", "Filtered_Multi_candidate_party_bar_color.png"),
  plot = p_party_color_multi_candidate,
  width = 6.5,
  height = 4.2,
  dpi = 300
)

ggsave(
  filename =
    here("output", "figures", "Filtered_Multi_candidate_party_bar_color.pdf"),
  plot = p_party_color_multi_candidate,
  width = 6.5,
  height = 4.2
)
p_party_color_multi_candidate
```



Most IMPORTANT peace of this Thesis, the Regression of the multi candidate model introducing candidate numbers as an additional factor.

Multi-candidate election: regression analysis This code estimates six OLS regression models for multi-candidate elections, using candidate vote share as the dependent variable. Models (1) and (2) use only candidates with observed individual RIS values, while Models (3) to (6) use the full sample and apply party-year ideology proxies where individual RIS values are missing. Models (5) and (6) extend the baseline specifications by adding the number of candidates, with Model (6) including an interaction between ideological distance and candidate number. Standard errors are clustered at the election level. The table also includes descriptive rows on centrist candidate success, average candidate numbers, and winner/loser vote shares. The final regression table is exported as a LaTeX file and can be loaded directly into Overleaf.

Standard errors are clustered at the election level because candidates competing in the same election are not independent observations. They share the same electoral context, and their vote shares are mechanically linked within each contest. Clustering therefore accounts for within-election correlation in the regression errors.

```
# ====================================================
# Multi-candidate elections: regression analysis
# ====================================================
```

```r
# --------------------------------------------------
# 0) Prepare analysis data
# --------------------------------------------------

Multi_candidate_ideology <- Multi_candidate_ideology |>
  group_by(election_id) |>
  mutate(
    n_candidates = n(),
    centrist_rank = min_rank(ideol_distance_0),
    most_centrist = if_else(centrist_rank == 1, 1, 0)
  ) |>
  ungroup()

# --------------------------------------------------
# 1) Regression models
# --------------------------------------------------

# RIS only
m1 <- lm(
  vote_share ~ ideol_distance_0,
  data = filter(Multi_candidate_ideology, ideology_source == "RIS")
)

m2 <- lm(
  vote_share ~ ideol_distance_0 + re_election_bin +
    female + age_election + party_affiliated,
  data = filter(Multi_candidate_ideology, ideology_source == "RIS")
)

# Full sample including party-year ideology proxies
m3 <- lm(
  vote_share ~ ideol_distance_0,
  data = Multi_candidate_ideology
)

m4 <- lm(
  vote_share ~ ideol_distance_0 + re_election_bin +
    female + age_election + party_affiliated,
  data = Multi_candidate_ideology
)

# Extended multi-candidate models
```

```r
m5 <- lm(
  vote_share ~ ideol_distance_0 + n_candidates +
    re_election_bin + female + age_election + party_affiliated,
  data = Multi_candidate_ideology
)

m6 <- lm(
  vote_share ~ ideol_distance_0 * n_candidates + re_election_bin +
    female + age_election + party_affiliated,
  data = Multi_candidate_ideology
)

# --------------------------------------------------
# 2) Cluster-robust VCOV matrices
# --------------------------------------------------

vcov_list <- list(
  vcovCL(m1, cluster = ~ election_id, type = "HC1"),
  vcovCL(m2, cluster = ~ election_id, type = "HC1"),
  vcovCL(m3, cluster = ~ election_id, type = "HC1"),
  vcovCL(m4, cluster = ~ election_id, type = "HC1"),
  vcovCL(m5, cluster = ~ election_id, type = "HC1"),
  vcovCL(m6, cluster = ~ election_id, type = "HC1")
)

# --------------------------------------------------
# 3) Descriptive statistics for table
# --------------------------------------------------

# Most centrist candidate wins
centrist_winrate <- Multi_candidate_ideology |>
  group_by(election_id) |>
  summarise(
    winner_rank = centrist_rank[win == 1][1],
    winner_is_most_centrist = winner_rank == 1,
    .groups = "drop"
  ) |>
  summarise(
    centrist_winrate = mean(winner_is_most_centrist, na.rm = TRUE)
  ) |>
  pull(centrist_winrate)
```

```r
# Winner among two most centrist candidates
top2_centrist_winrate <- Multi_candidate_ideology |>
  group_by(election_id) |>
  summarise(
    winner_rank = centrist_rank[win == 1][1],
    winner_top2_centrist = winner_rank <= 2,
    .groups = "drop"
  ) |>
  summarise(
    top2_centrist_winrate = mean(winner_top2_centrist, na.rm = TRUE)
  ) |>
  pull(top2_centrist_winrate)

# Mean number of candidates per election
mean_n_candidates <- Multi_candidate_ideology |>
  distinct(election_id, n_candidates) |>
  summarise(mean_n_candidates = mean(n_candidates, na.rm = TRUE)) |>
  pull(mean_n_candidates)

# Winner/loser average vote shares
vote_share_stats <- Multi_candidate_ideology |>
  mutate(status = if_else(win == 1, "Winner", "Loser")) |>
  group_by(status) |>
  summarise(
    mean_vote_share = mean(vote_share, na.rm = TRUE),
    .groups = "drop"
  )

winner_mean <- vote_share_stats |>
  filter(status == "Winner") |>
  pull(mean_vote_share)

loser_mean <- vote_share_stats |>
  filter(status == "Loser") |>
  pull(mean_vote_share)

# --------------------------------------------------
# 4) Extra rows for regression table
# --------------------------------------------------

extra_rows_multi <- tribble(
  ~term, ~`(1)`, ~`(2)`, ~`(3)`, ~`(4)`, ~`(5)`, ~`(6)`,
```

```r
  "Most centrist candidate wins",
  sprintf("%.0f%%", 100 * centrist_winrate), "", "", "", "", "",
  "Winner among two most centrist",
  sprintf("%.0f%%", 100 * top2_centrist_winrate), "", "", "", "", "",
  "Mean no. of candidates",
  sprintf("%.2f", mean_n_candidates), "", "", "", "", "",
  "Mean vote share, winners",
  sprintf("%.3f", winner_mean), "", "", "", "", "",
  "Mean vote share, losers",
  sprintf("%.3f", loser_mean), "", "", "", "", ""
)

# --------------------------------------------------
# 5) Nice coefficient names
# --------------------------------------------------

coef_names <- c(
  "(Intercept)" = "Constant",
  "ideol_distance_0" = "Ideological distance",
  "re_election_bin" = "Incumbent",
  "female" = "Female",
  "age_election" = "Age",
  "party_affiliated" = "Party affiliated",
  "n_candidates" = "Number of candidates",
  "ideol_distance_0:n_candidates" =
    "Ideological distance $\\times$ number of candidates"
)

gof_map_multi <- tribble(
  ~raw, ~clean, ~fmt,
  "nobs", "Observations", 0,
  "r.squared", "R-squared", 3
)

# --------------------------------------------------
# 6) Export LaTeX table for Overleaf
# --------------------------------------------------

modelsummary(
  list(
    "(1)" = m1,
    "(2)" = m2,
```

```
    "(3)" = m3,
    "(4)" = m4,
    "(5)" = m5,
    "(6)" = m6
  ),
  coef_map = coef_names,
  vcov = vcov_list,
  statistic = "({std.error})",
  stars = c('*' = .1, '**' = .05, '***' = .01),
  gof_map = gof_map_multi,
  add_rows = extra_rows_multi,
  title = "Multi-candidate elections: ideological distance and vote share",
  notes = c(
    "Dependent variable: candidate vote share.",
    "Models (1) and (2) include only candidates with observed RIS values.",
    "Models (3) to (6) use the full sample and include
    #party-year ideology proxies where RIS is unavailable.",
    "Standard errors clustered at the election level
  #are reported in parentheses.",
    "Models (5) and (6) account explicitly for
    Çthe number of candidates in the election."
  ),
  escape = FALSE,
  output = here("output", "tables", "tab_multi_candidate_regressions.tex")
)
```

Additional clarifications regarding the table caption and explanatory notes were added directly to the LaTeX table output in Overleaf.

6 Visualization Voter turnout

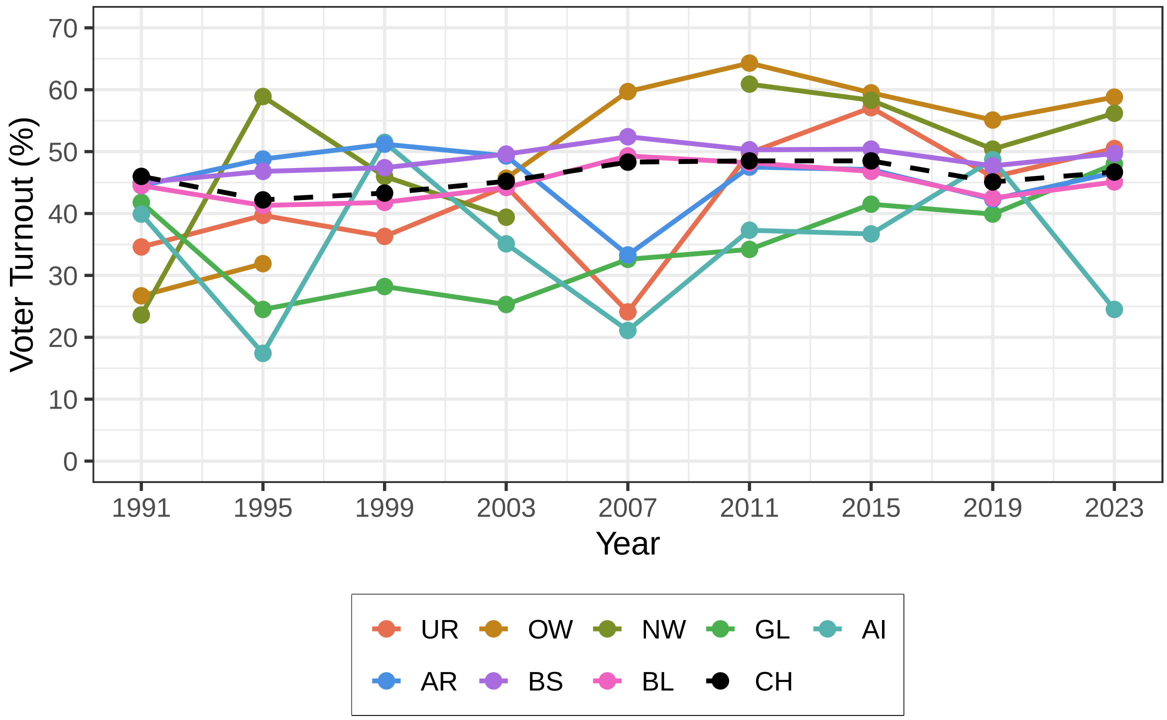
Loading the manually transcribed BFS voter turnout data. Transferring the data into a separate file was necessary because the original source was formatted for presentation rather than data analysis and was therefore not directly compatible with the required processing structure.

```
# 1) Load data
voter_turnout_single_seat_cantons <- read_csv(
  here("data", "raw", "voter_turnout_single_seat_cantons.csv"),
  show_col_types = FALSE
)
```

```r
voter_turnout_single_seat_cantons_long <- voter_turnout_single_seat_cantons |>
  pivot_longer(-Year, names_to = "canton", values_to = "turnout_raw") |>
  mutate(
    turnout = readr::parse_number(turnout_raw),
    canton = factor(
      canton,
      levels = c(
        "Uri","Obwalden","Nidwalden","Glarus",
        "Appenzell Innerrhoden","Appenzell Ausserrhoden",
        "Basel-Stadt","Basel-Landschaft","Schweiz"
      )
    ),
    is_ch = canton == "Schweiz"
  )
p_turnout <- ggplot(voter_turnout_single_seat_cantons_long,
                    aes(x = Year, y = turnout,
                                 colour = canton,
                                 linetype = canton)) +
  geom_line(linewidth = 1.1) +
  geom_point(size = 3) +
  scale_x_continuous(breaks =
              sort(unique(voter_turnout_single_seat_cantons_long$Year))) +
  scale_y_continuous(limits = c(0, 70), breaks = seq(0, 70, 10)) +

  scale_colour_manual(
    values = c(
      "Uri" = "#E76F51",
      "Obwalden" = "#C0841A",
      "Nidwalden" = "#7A8F28",
      "Glarus" = "#4CAF50",
      "Appenzell Innerrhoden" = "#55B2AE",
      "Appenzell Ausserrhoden" = "#4A90E2",
      "Basel-Stadt" = "#A86BE0",
      "Basel-Landschaft" = "#F062C0",
      "Schweiz" = "black"
    ),
    labels = c(
      "Uri" = "UR",
      "Obwalden" = "OW",
      "Nidwalden" = "NW",
      "Glarus" = "GL",
      "Appenzell Innerrhoden" = "AI",
```

```
    "Appenzell Ausserrhoden" = "AR",
    "Basel-Stadt" = "BS",
    "Basel-Landschaft" = "BL",
    "Schweiz" = "CH"
  )
) +

scale_linetype_manual(
  values = c(
    "Uri" = "solid",
    "Obwalden" = "solid",
    "Nidwalden" = "solid",
    "Glarus" = "solid",
    "Appenzell Innerrhoden" = "solid",
    "Appenzell Ausserrhoden" = "solid",
    "Basel-Stadt" = "solid",
    "Basel-Landschaft" = "solid",
    "Schweiz" = "dashed"
  ),
  labels = c(
    "Uri" = "UR",
    "Obwalden" = "OW",
    "Nidwalden" = "NW",
    "Glarus" = "GL",
    "Appenzell Innerrhoden" = "AI",
    "Appenzell Ausserrhoden" = "AR",
    "Basel-Stadt" = "BS",
    "Basel-Landschaft" = "BL",
    "Schweiz" = "CH"
  )
) +

labs(x = "Year", y = "Voter Turnout (%)") +
theme_bw(base_size = 16) +
theme(
  legend.position = "bottom",
  legend.title = element_blank(),
  legend.box.background = element_rect(color = "black"),
  legend.key = element_blank()
) +
guides(
  colour = guide_legend(nrow = 2, byrow = TRUE)
```

```
  )

p_turnout
```




```
# 4) Save (use your path conventions if you want)
ggsave(
  filename = here("output", "figures", "Voter_Turnout_overworked.png"),
  plot = p_turnout,
  width = 6.5,
  height = 4.2,
  dpi = 300
)
ggsave(
  filename = here("output", "figures", "Voter_Turnout_overworked.pdf"),
  plot = p_turnout,
  width = 6.5,
  height = 4.2
)
```

```
write_csv(
  voter_turnout_single_seat_cantons_long,
  here("data", "clean", "voter_turnout_single_seat_cantons_long.csv")
)
saveRDS(
  voter_turnout_single_seat_cantons_long,
  here("data", "clean", "voter_turnout_single_seat_cantons_long.rds")
)
```

Furthermore, this chunk checks the influence of the amount of Candidates (After filtering for others) in NC elections on the Voter Turnout. This additional analysis examines whether candidate supply is associated with NC voter turnout in single-seat cantons. The sample starts in 2003, corresponding to the first election year covered by the collected candidate dataset. Since the turnout measure refers to NC elections, Basel-Stadt and Basel-Landschaft are excluded because they hold more than one NC seat. Appenzell Innerrhoden is also excluded due to its Landsgemeinde voting system. The first specification uses only the number of NC candidates, while the second specification uses the combined number of National Council and Council of States candidates, since both elections often take place on the same day.

```
# ==================================================
# Complex Candidate numbers and NC voter turnout
# ==================================================

# --------------------------------------------------
# 1) Define relevant NC single-seat cantons
# --------------------------------------------------
# Appenzell Innerrhoden is excluded because of the Landsgemeinde system.
# Basel-Stadt and Basel-Landschaft are excluded because they have more
# than one National Council seat.

nc_single_seat_cantons <- c(
  "Uri",
  "Obwalden",
  "Nidwalden",
  "Glarus",
  "Appenzell Ausserrhoden"
)

# --------------------------------------------------
# 2) Load data
# --------------------------------------------------
```

```r
turnout <- voter_turnout_single_seat_cantons_long

cand_turnout <- cand_real

# --------------------------------------------------
# 3) Prepare voter turnout data
# --------------------------------------------------
# Turnout data refer to National Council elections.
# The analysis starts in 2003, because this is the first election year
# covered by the collected candidate dataset.

turnout_analysis <- turnout |>
  filter(canton %in% nc_single_seat_cantons) |>
  rename(
    year = Year,
    turnout_nc = turnout
  ) |>
  filter(year >= 2003) |>
  select(year, canton, turnout_nc)

# --------------------------------------------------
# 4) Count candidates by canton-year
# --------------------------------------------------
# cand_real already excludes "Others".
# We count National Council candidates only and then National Council
# plus Council of States candidates combined.

candidate_counts <- cand_turnout |>
  filter(canton %in% nc_single_seat_cantons) |>
  filter(year >= 2003) |>
  group_by(year, canton) |>
  summarise(
    n_nc_candidates = sum(chamber == "NR" | chamber == "NC"),
    n_cs_candidates = sum(chamber == "SR" | chamber == "CS"),
    n_total_candidates = n_nc_candidates + n_cs_candidates,
    .groups = "drop"
  )

# --------------------------------------------------
# 5) Merge turnout and candidate counts
# --------------------------------------------------
```

```r
turnout_reg_data <- turnout_analysis |>
  left_join(candidate_counts, by = c("year", "canton")) |>
  mutate(
    n_nc_candidates = replace_na(n_nc_candidates, 0),
    n_cs_candidates = replace_na(n_cs_candidates, 0),
    n_total_candidates = replace_na(n_total_candidates, 0)
  )

print(turnout_reg_data)
```

```
# A tibble: 30 x 6
    year canton    turnout_nc n_nc_candidates n_cs_candidates n_total_candidates
   <dbl> <chr>          <dbl>           <int>           <int>              <int>
 1  2023 Uri             50.5               2               0                  2
 2  2023 Obwalden        58.8               2               1                  3
 3  2023 Nidwalden       56.2               3               3                  6
 4  2023 Glarus          48                 4               0                  4
 5  2023 Appenzel~       46.6               3               2                  5
 6  2019 Uri             45.9               3               0                  3
 7  2019 Obwalden        55.1               5               1                  6
 8  2019 Nidwalden       50.4               2               1                  3
 9  2019 Glarus          39.9               3               0                  3
10  2019 Appenzel~       42.3               2               2                  4
# i 20 more rows
```

```r
write_csv(
  turnout_reg_data,
  here("data", "clean", "turnout_candidate_regression_data.csv")
)

# ---------------------------------------------------
# 6) Linear regression models
# ---------------------------------------------------

# Model 1: only National Council candidates
m_turnout_nc <- lm(
  turnout_nc ~ n_nc_candidates,
  data = turnout_reg_data
)

# Model 2: National Council and Council of States candidates combined
```

```r
m_turnout_total <- lm(
  turnout_nc ~ n_total_candidates,
  data = turnout_reg_data
)

# Model 3: National Council and Council of States candidates separately
m_turnout_split <- lm(
  turnout_nc ~ n_nc_candidates + n_cs_candidates,
  data = turnout_reg_data
)

m_turnout_fe <- lm(
  turnout_nc ~ n_total_candidates + factor(canton) + factor(year),
  data = turnout_reg_data
)

# --------------------------------------------------
# 7) Regression table
# --------------------------------------------------

modelsummary(
  list(
    "NC candidates only" = m_turnout_nc,
    "NC + CS candidates" = m_turnout_total,
    "NC and CS separately" = m_turnout_split,
    "Canton/year FE" = m_turnout_fe
  ),
  stars = TRUE,
  statistic = "std.error",
  gof_omit = "IC|Log|F",
  output = here("output", "tables", "turnout_candidate_regression.tex")
)
```

The extended specifications shown above were estimated as an exploratory attempt to account for the broader electoral context in which NC turnout is observed. However, given the small number of observations and the fact that NC elections often coincide with CS elections and other national votes, the resulting models are difficult to interpret in a meaningful way. In particular, the inclusion of additional candidate counts and fixed effects substantially increases model complexity without providing a clear substantive insight.

For this reason, the analysis is reduced to a simpler bivariate specification focusing only on the association between the number of NC candidates and NC voter turnout. This specification should not be interpreted causally. Rather, it serves as a descriptive robustness check assessing whether there is any basic relationship between candidate supply and turnout in the available single-seat canton sample. Important contextual factors, such as concurrent CS elections, national votes held on the same day, local mobilization, and canton-specific political traditions, remain outside the model and limit the interpretability of the result.

```
# ==================================================
# Simple Candidate numbers and NC voter turnout
# ==================================================
# --------------------------------------------------
# 1) Define relevant NC single-seat cantons
# --------------------------------------------------

nc_single_seat_cantons <- c(
  "Uri",
  "Obwalden",
  "Nidwalden",
  "Glarus",
  "Appenzell Ausserrhoden"
)

# --------------------------------------------------
# 2) Load data
# --------------------------------------------------

turnout <- voter_turnout_single_seat_cantons_long

cand_turnout <- cand_real

# --------------------------------------------------
# 3) Prepare voter turnout data
# --------------------------------------------------
# Turnout data refer to National Council elections.
# The analysis starts in 2003, because this is the first election year
# covered by the collected candidate dataset.

turnout_analysis <- turnout |>
  filter(canton %in% nc_single_seat_cantons) |>
  rename(
    year = Year,
    turnout_nc = turnout
  ) |>
  filter(year >= 2003) |>
```

```r
  select(year, canton, turnout_nc)

# ---------------------------------------------------
# 4) Count National Council candidates
# ---------------------------------------------------
# cand_real already excludes the aggregated residual category
# "Others". Elections with only one identifiable National Council candidate
# are excluded from this additional turnout regression, because some of these
# cases may still contain vote shares for aggregated "Others" categories.
# Including them would understate the actual number of alternatives available
# to voters.

candidate_counts_nc <- cand |>
  filter(canton %in% nc_single_seat_cantons) |>
  filter(year >= 2003) |>
  group_by(year, canton) |>
  summarise(
    n_nc_candidates = sum(chamber == "NR" | chamber == "NC"),
    .groups = "drop"
  ) |>
  filter(n_nc_candidates > 1)

# ---------------------------------------------------
# 5) Merge turnout and candidate counts
# ---------------------------------------------------
# An inner join is used so that only canton-year observations with a valid
# candidate count after the above restrictions are retained.

turnout_reg_data <- turnout_analysis |>
  inner_join(candidate_counts_nc, by = c("year", "canton"))

# Inspect final regression dataset
print(turnout_reg_data)
```

```
# A tibble: 29 x 4
    year canton                 turnout_nc n_nc_candidates
   <dbl> <chr>                       <dbl>           <int>
 1  2023 Uri                          50.5               2
 2  2023 Obwalden                     58.8               2
 3  2023 Nidwalden                    56.2               3
 4  2023 Glarus                       48                 4
 5  2023 Appenzell Ausserrhoden       46.6               3
```

```
 6  2019 Uri                           45.9              4
 7  2019 Obwalden                      55.1              5
 8  2019 Nidwalden                     50.4              2
 9  2019 Glarus                        39.9              4
10  2019 Appenzell Ausserrhoden        42.3              3
# i 19 more rows
```

```r
# Optional: save regression dataset
write_csv(
  turnout_reg_data,
  here("data", "clean", "turnout_candidate_regression_data.csv")
)

# --------------------------------------------------
# 6) Simple linear regression
# --------------------------------------------------
# The model estimates the association between the number of identifiable
# National Council candidates and National Council voter turnout.

m_turnout_nc_simple <- lm(
  turnout_nc ~ n_nc_candidates,
  data = turnout_reg_data
)

summary(m_turnout_nc_simple)
```

```
Call:
lm(formula = turnout_nc ~ n_nc_candidates, data = turnout_reg_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-27.1680  -5.9556   0.0444   7.5320  16.5505 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)       56.680      5.610    10.1 1.14e-10 ***
n_nc_candidates   -2.706      1.503    -1.8    0.083 .  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.11 on 27 degrees of freedom
```

```
Multiple R-squared:  0.1072,    Adjusted R-squared:  0.07411
F-statistic: 3.241 on 1 and 27 DF,  p-value: 0.08299
```

```
# --------------------------------------------------
# 7) Export small regression table
# --------------------------------------------------

modelsummary(
  list(
    "NC turnout" = m_turnout_nc_simple
  ),
  stars = TRUE,
  statistic = "std.error",
  gof_omit = "IC|Log|F",
  coef_rename = c(
    "(Intercept)" = "Constant",
    "n_nc_candidates" = "Number of NC candidates"
  ),
  output = here("output", "tables", "turnout_nc_candidates_simple.tex")
)
```

7 Further Analysis: Multi and Two candidate combined analysis

The main analysis separates two-candidate and multi-candidate elections, as the theoretical mechanisms differ across these electoral settings. This additional dataset pools both types of competitive elections only for exploratory graphing and robustness checks. It is not used as a main empirical specification in the thesis.

```
Plural_candidate_elections <- cand_analysis_clean |>
  filter(regime %in% c("Two-candidate", "Multi-candidate"))

write_csv(
  Plural_candidate_elections,
  here("data", "clean", "Plural_candidate_elections.csv")
)
saveRDS(
  Plural_candidate_elections,
  here("data", "clean", "Plural_candidate_elections.rds")
)
```

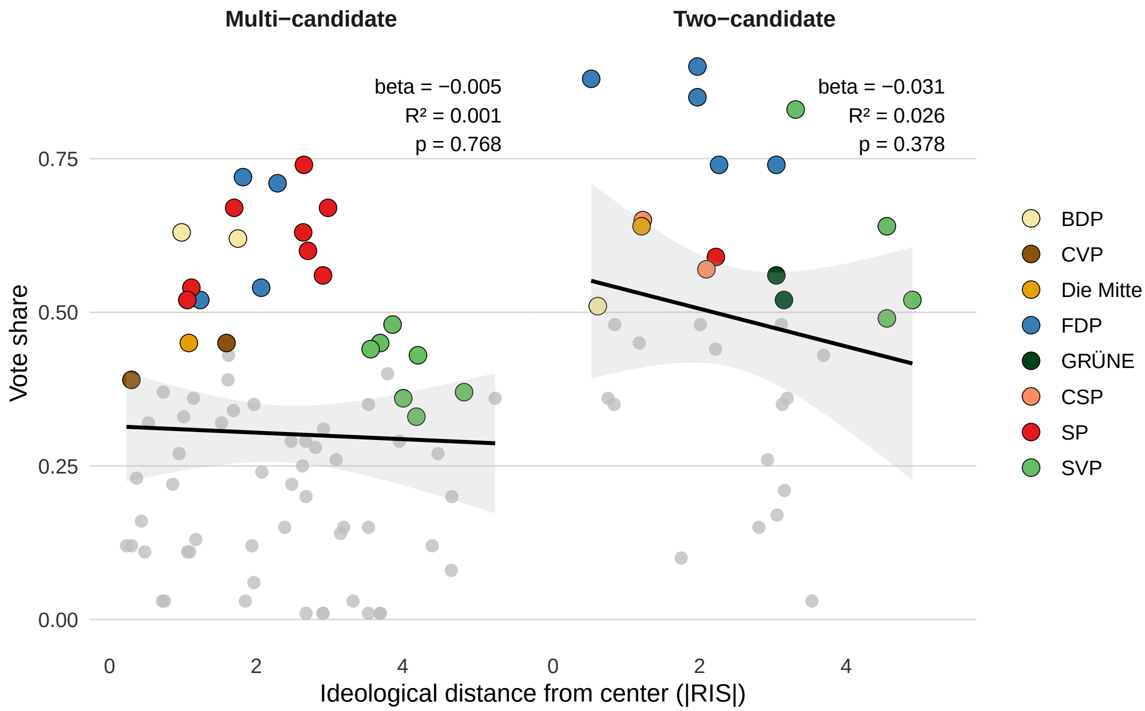
```r
Plural_candidate_ideology <- Plural_candidate_elections |>
  mutate(
    # use observed RIS if available, otherwise the party-year proxy
    ris_used = coalesce(ris, as.numeric(pol_party_value)),

    # flag which source was used
    ideology_source = case_when(
      !is.na(ris) ~ "RIS",
      is.na(ris) & !is.na(pol_party_value) ~ "Party proxy",
      TRUE ~ "Missing"
    ),
# Adjust the relection bin so that all
# non reelected are 0 not just not reelected winners
      re_election_bin = case_when(
      re_election == "Yes" ~ 1,
      TRUE ~ 0
    ),

    # ideological distance from assumed center / proxy median voter at 0
    ideol_distance_0 = abs(ris_used)
  ) |>
  filter(!is.na(ris_used))
```

```r
Plural_candidate_plot <- Plural_candidate_ideology |>
  mutate(
    regime = factor(
      regime,
      levels = c("Multi-candidate", "Two-candidate")
    ),
    plot_colour = if_else(win == 1, party, "Loser"),
    plot_colour = factor(
      plot_colour,
      levels = c(
        "BDP", "CVP", "Die Mitte", "FDP", "GLP", "GRÜNE",
        "CSP", "JUSO", "LPS", "SP", "SVP", "Loser"
      )
    )
  )

plot_colors <- c(
  party_colors,
  "Loser" = "grey75"
```

```
)

regression_labels <- Plural_candidate_plot |>
  group_by(regime) |>
  summarise(
    beta = coef(lm(vote_share
                  ~ ideol_distance_0, data = cur_data()))[2],
    r2   = summary(lm(vote_share
                     ~ ideol_distance_0, data = cur_data()))$r.squared,
    p    = coef(summary(lm(vote_share
                          ~ ideol_distance_0, data = cur_data())))[2, 4],
    .groups = "drop"
  ) |>
  mutate(
    label = paste0(
      "beta = ", sprintf("%.3f", beta),
      "\nR² = ", sprintf("%.3f", r2),
      "\np = ", ifelse(p < 0.001, "< 0.001", sprintf("%.3f", p))
    ),
    x = 5.35,
    y = 0.88
  )

PlotPluralcandidate_Faceted_Voteshare_RIS <- ggplot(
  Plural_candidate_plot,
  aes(x = ideol_distance_0, y = vote_share)
) +
  geom_point(
    data = \(d) d |> dplyr::filter(win == 0),
    color = "grey75",
    size = 2.3,
    alpha = 0.8
  ) +
  geom_point(
    data = \(d) d |> dplyr::filter(win == 1),
    aes(fill = plot_colour),
    shape = 21,
    color = "black",
    stroke = 0.3,
    size = 3.8
  ) +
  geom_smooth(
```

```
    method = "lm",
    se = TRUE,
    color = "black",
    linewidth = 0.9,
    fill = "grey70",
    alpha = 0.22
  ) +
  geom_text(
    data = regression_labels,
    aes(x = x, y = y, label = label),
    inherit.aes = FALSE,
    hjust = 1,
    vjust = 1,
    size = 3.4
  ) +
  facet_wrap(~ regime, nrow = 1) +
  scale_fill_manual(
    values = plot_colors,
    breaks = c(
      "BDP", "CVP", "Die Mitte", "FDP", "GLP", "GRÜNE",
      "CSP", "JUSO", "LPS", "SP", "SVP"
    )
  ) +
  coord_cartesian(
    xlim = c(0, 5.5),
    ylim = c(0, 0.9)
  ) +
  labs(
    x = "Ideological distance from center (|RIS|)",
    y = "Vote share",
    fill = NULL
  ) +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "right",
    legend.text = element_text(size = 10),
    strip.text = element_text(size = 11, face = "bold"),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    panel.grid.major.y = element_line(color = "grey85", linewidth = 0.35),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10, color = "grey20"),
```

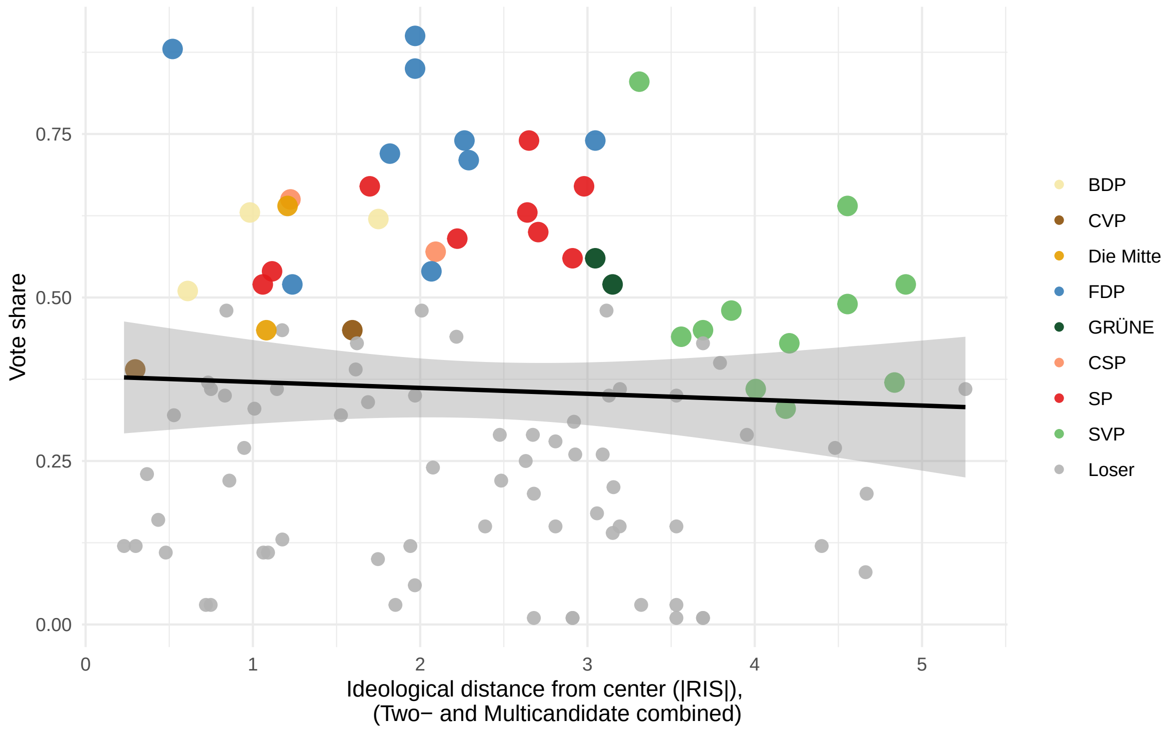
```
    plot.margin = margin(8, 12, 8, 8),
    panel.spacing = grid::unit(0, "pt")
  )

ggsave(
  filename =
    here("output", "figures", "Pluralcandidate_facet_voteshare_RIS.png"),
  plot = PlotPluralcandidate_Faceted_Voteshare_RIS,
  width = 8.5,
  height = 4.2,
  dpi = 300
)

ggsave(
  filename =
    here("output", "figures", "Pluralcandidate_facet_voteshare_RIS.pdf"),
  plot = PlotPluralcandidate_Faceted_Voteshare_RIS,
  width = 8.5,
  height = 4.2
)

PlotPluralcandidate_Faceted_Voteshare_RIS
```



The bivariate relationship between ideological distance from the center and vote share is weak in both electoral settings. In multi-candidate elections, the estimated relationship is essentially flat. In two-candidate elections, the slope is more negative, but the association remains statistically imprecise and explains little of the overall variation in vote share. This plot shows the combined outcome of the plural candidate dataset.

```
Combined_candidate_plot <- Plural_candidate_ideology |>
  mutate(
    plot_colour = if_else(win == 1, party, "Loser")
  )
plot_colors <- c(party_colors, "Loser" = "grey70")
## PLOT with Party colors as the colors for won ellections
PlotCombinedcandidate_Voteshare_RIS <-
Combined_candidate_plot |>
  mutate(
    plot_colour = factor(plot_colour,
                         levels = c(
                           "BDP","CVP","Die Mitte",
                           "FDP","GLP","GRÜNE",
                           "CSP","JUSO","LPS","SP","SVP","Loser"
                         ))
```

```r
  ) |>
  ggplot(aes(x = ideol_distance_0, y = vote_share)) +
  geom_point(aes(color = plot_colour, size = factor(win)), alpha = 0.9) +
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  scale_color_manual(values = plot_colors) +
  scale_size_manual(values = c("0" = 2.5, "1" = 4), guide = "none") +
  labs(
    x = "Ideological distance from center (|RIS|),
    (Two- and Multicandidate combined)",
    y = "Vote share",
    color = NULL
  ) +
  theme_minimal()
ggsave(
  filename = here("output", "figures", "Combinedcandidate_voteshare_RIS.png"),
  plot = PlotCombinedcandidate_Voteshare_RIS,
  width = 6.5,
  height = 4.2,
  dpi = 300
)

ggsave(
  filename = here("output", "figures", "Combinedcandidate_voteshare_RIS.pdf"),
  plot = PlotCombinedcandidate_Voteshare_RIS,
  width = 6.5,
  height = 4.2
)
PlotCombinedcandidate_Voteshare_RIS
```

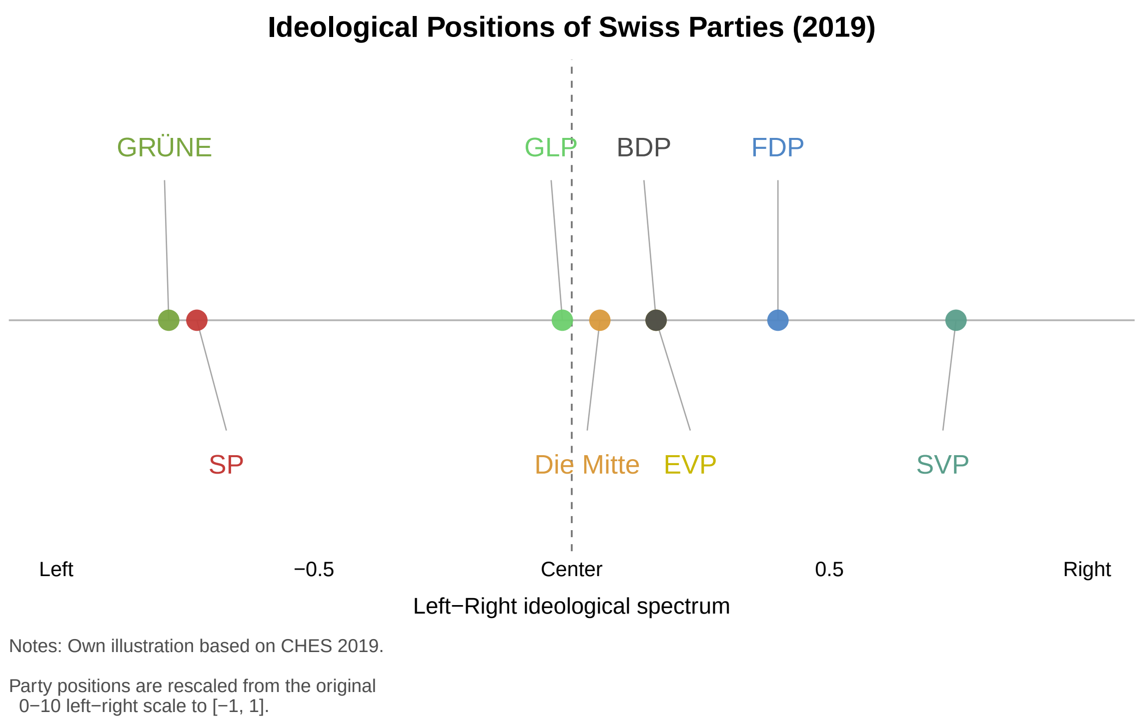


APPENDIX B (From Thesis)

```
# Load CHES 2019 data
CHES19_DF_FULL <- read_csv(here("data", "raw", "CHES2019V3.csv"))

# Filter Switzerland
CHES19_CH <- CHES19_DF_FULL %>%
  filter(country == 36)

# Scale ideology variable and harmonize party names
CHES19_Scaled <- CHES19_CH %>%
  mutate(lrgen_scaled = (lrgen - 5) / 5) %>%
  select(party, lrgen, lrgen_scaled) %>%
  mutate(
    party = recode(
      party,
      "SVP/UDC" = "SVP",
      "SP/PS" = "SP",
      "FDP/PLR" = "FDP",
```

```r
      "CVP/PVC" = "Die Mitte",
      "GPS/PES" = "GRÜNE",
      "GPL/PVL" = "GLP",
      "EVP/PEV" = "EVP",
      "BDP/PBD" = "BDP"
    )
  )

# Thesis-friendly colors
party_colors <- c(
  "SVP" = "#5A9E8B",
  "SP" = "#C43C39",
  "FDP" = "#4F86C6",
  "Die Mitte" = "#D99A3D",
  "GRÜNE" = "#7AA641",
  "GLP" = "#6CCF6C",
  "EVP" = "#C9B800",
  "BDP" = "#4D4D4D"
)


label_df <- tibble(
  party = c("GRÜNE", "SP", "GLP", "Die Mitte", "BDP", "EVP", "FDP", "SVP"),
  x_lab = c(-0.79, -0.67, -0.04, 0.03, 0.14, 0.23, 0.40, 0.72),
  y_lab = c( 0.055, -0.045, 0.055, -0.045, 0.055, -0.045, 0.055, -0.045),
  vjust = c(0, 1, 0, 1, 0, 1, 0, 1)
)

plot_df <- CHES19_Scaled %>%
  left_join(label_df, by = "party")

 CHES_partydistribution<- ggplot(plot_df,
                                 aes(x = lrgen_scaled, y = 0,
                                     color = party)) +
  geom_hline(yintercept = 0, color = "gray70", linewidth = 0.4) +
  geom_vline(xintercept = 0, linetype = "dashed", color
             = "gray45", linewidth = 0.4) +
  geom_segment(
    aes(x = lrgen_scaled, xend = x_lab,
        y = 0, yend = y_lab - ifelse(y_lab > 0, 0.008, -0.008)),
    color = "gray65",
    linewidth = 0.3,
```

```
    inherit.aes = FALSE
  ) +
  geom_point(size = 4.2, alpha = 0.95) +
  geom_text(
    aes(x = x_lab, y = y_lab, label = party, vjust = vjust),
    size = 4.6,
    show.legend = FALSE
  ) +
  scale_color_manual(values = party_colors) +
  scale_x_continuous(
    limits = c(-1.05, 1.05),
    breaks = c(-1, -0.5, 0, 0.5, 1),
    labels = c("Left", "-0.5", "Center", "0.5", "Right"),
    expand = expansion(mult = c(0.02, 0.02))
  ) +
  coord_cartesian(ylim = c(-0.07, 0.08), clip = "off") +
 labs(
  title = "Ideological Positions of Swiss Parties (2019)",
  x = "Left-Right ideological spectrum",
  y = NULL,
  caption = "Notes: Own illustration based on CHES 2019.
  \nParty positions are rescaled from the original
  0-10 left-right scale to [-1, 1]."
)+
  theme_minimal(base_size = 11) +
  theme(
    legend.position = "none",
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank(),
    panel.grid.major.y = element_blank(),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.title.x = element_text(size = 11, margin = margin(t = 8)),
    axis.text.x = element_text(size = 10, color = "black"),
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14, margin = margin(b = 8))
    plot.caption = element_text(hjust = 0, size = 9, color = "gray30", margin = margin(t = 1
    plot.margin = margin(10, 15, 10, 15)
  )
 CHES_partydistribution
```




```r
ggsave(
  filename = here("output", "figures", " CHESAppendixB.png"),
  plot =  CHES_partydistribution,
  width = 6.5,
  height = 4.2,
  dpi = 300
)

ggsave(
  filename = here("output", "figures", " CHESAppendixB.pdf"),
  plot =  CHES_partydistribution,
  width = 6.5,
  height = 4.2
)
```